High-Order Unconditionally Stable IMEX BDF–FEM Schemes for the Hodgkin–Huxley Model on a Branched Axon

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Abstract

We construct and analyze high-order implicit–explicit backward differentiation formula finite element (IMEX BDF–FEM) schemes for the Hodgkin–Huxley cable system on a branched metric graph. The linearly implicit schemes combine constant-step BDF methods of orders $q = 1, \dots, 5$ with conforming finite elements of any fixed degree $p \geq 1$. Extrapolated gate variables are smoothly masked before evaluating the conductances, preserving coercivity without projecting the numerical gates. A voltage cutoff makes the gate rates globally bounded and Lipschitz. Each timestep requires one linear voltage solve followed by independent nodal gate updates. Under exact spatial integration, we prove unconditional stability of the complete fully discrete regularized scheme in discrete history energies. Under regularity and starting-history assumptions, the voltage and reconstructed gates converge in $L^2(\Gamma)$ at the optimal rate $O(h^{p+1} + \Delta t^q)$, without a timestep restriction or an h–Δt coupling. The same rate holds with numerical quadrature when its full residual defect is $O(h^{p+1})$. Manufactured solutions recover the predicted orders through BDF5–P_4. Action-potential simulations illustrate pulse propagation and splitting, while a controlled work–precision comparison within the proposed family shows that higher-order variants can reach a prescribed accuracy at much lower runtime along the tested joint-refinement paths.

Keywords: Hodgkin–Huxley model, metric graph, finite element method, implicit–explicit schemes, unconditional stability, optimal convergence rate
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1. Introduction

The Hodgkin–Huxley cable model describes action-potential propagation by coupling a parabolic voltage equation to three ionic gate equations [1]. A branched axon is naturally represented as a metric graph: each edge is a cable segment, and voltage continuity and Kirchhoff current balance couple the branches at their junctions. Rapid voltage fronts and distinct gate time scales make the system stiff. Accurate simulation therefore requires spatial resolution together with stable time integration and error control for the coupled system.

Neuronal methods include branched-cable solvers [2], exponential gate updates [3], and first- and second-order exponential integrators that prevent voltage overshoot for space-clamped Hodgkin–Huxley-like ODEs [4]. For the cable PDE–ODE system, Mascagni [5] proved an $O(\Delta t + h^2)$ error

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bound using backward Euler in time and centered finite differences in space. Other cable discretizations address high-order spatial accuracy [6], positivity and boundedness [7], and charge conservation [8]. Branchwise predictor–corrector methods decouple cable segments at junctions [9]; a locally adaptive BDF2 finite-difference method exhibits second-order convergence in numerical tests [10]. Amsallem and Nordström [11] established well-posedness for the continuous problem and semi-discrete energy estimates for high-order SBP–SAT schemes on branched Hodgkin–Huxley cables.

Related neuronal calcium models also require a numerical treatment of coupled spatial fields and channel dynamics. Guan and Queisser [12] proved well-posedness and analyzed an IMEX finite element scheme for reaction–diffusion PDEs coupled to interface ODEs. Gurung and Guan [13] developed a hybrid approach in which a neural-network model of channel open probabilities is coupled to the calcium PDEs.

In cardiac electrophysiology, Africa et al. combine high-order Q_p spectral elements with BDF1–3 and a linear voltage solve [14]. A recent neuronal tissue DG method uses Crank–Nicolson time stepping and includes semi-discrete analysis for a simplified ionic problem [15].

Broader electrophysiology research addresses high-order and adaptive spatial approximation [16, 17, 18], quadrature and ionic interpolation [19], and time integration, splitting, and multirate or parallel-in-time methods [20, 21, 22, 23]. For FitzHugh–Nagumo systems, Cai et al. [24] proved first-order-in-time IMEX finite element error bounds. Chrysafinos et al. [25] analyzed discontinuous Galerkin time stepping with conforming spatial finite elements and proved arbitrary-order space–time error estimates for a system of two reaction–diffusion PDEs. Variable-step BDF2 finite element analysis assumes a bound on adjacent step ratios [26]. An unconditional invariant-set-preserving finite-difference scheme is also available [27].

In this paper, we construct and analyze constant-step IMEX BDF–finite element schemes for the Hodgkin–Huxley model on a branched metric graph. Our discretization builds on finite element methods for metric graphs [28, 29] and IMEX multistep analysis for parabolic equations [30, 31, 32, 33]. High-order extrapolation can move gate values outside $[0, 1]$ and destroy voltage-equation coercivity, while the classical gate rates are not globally bounded or Lipschitz in voltage. We apply a smooth mask to the extrapolated gates and a smooth cutoff to the voltage entering the rates. The mask preserves a positive total conductance, while the cutoff gives bounded Lipschitz rates. The mask is the identity on $[0, 1]$, and the cutoff is the identity on a prescribed voltage interval $[-M, M]$ containing the exact solution range. The numerical gates remain unprojected. To our knowledge, unconditional fully discrete energy bounds for regularized, linearly implicit BDF schemes of orders three through five for the coupled Hodgkin–Huxley cable–gate system on branched axons have not previously been established.

Our main contributions are:

1. We construct linearly implicit BDFq–P_p schemes for $q = 1, \ldots, 5$ and any fixed $p \geq 1$, with one linear voltage solve and independent nodal gate updates per timestep, without nonlinear iteration.
2. Using the gate mask and voltage cutoff, we prove unconditional stability in discrete history energies for the complete fully discrete BDF1–BDF5 schemes under exact spatial integration.
3. We prove optimal fully discrete voltage and gate error bounds under regularity and starting-history assumptions, and recover the same rate with numerical quadrature when its full residual defect is $O(h^{p+1})$.

4. We assess accuracy and efficiency through manufactured solutions and action-potential simulations.

The stability proof combines BDF multiplier estimates [34, 35, 36] with a positive nodal inner product for the gate equations. Let $\mathcal{W} = \{m, h, n\}$ denote the gates, I_h the degree-p nodal interpolation operator, and $\|\cdot\|_h$ the positive weighted norm of gate values at the Gauss–Lobatto nodes. Writing $\|\cdot\|$ for the $L^2(\Gamma)$ norm, the convergence estimate under exact integration and the stated regularity and starting-history assumptions takes the form

$$\max_n \left(\|V_h^n - V(t_n)\| + \sum_{w \in \mathcal{W}} \|w_h^n - I_h w(t_n)\|_h \right) \le C_{p,q}(T)\big(h^{p+1} + \Delta t^q\big),$$

without a timestep restriction or an h–Δt coupling. The nodal gate norm equivalence and interpolation give the same rate for the reconstructed gates in $L^2(\Gamma)$. The voltage update uses previous gate values and the gate update uses the new voltage, so their error bounds are combined sequentially.

Manufactured solutions recover the predicted orders through BDF5–P_4 on the Y-shaped graph; a further test checks spatial accuracy on a two-junction graph with unequal edge lengths. Action-potential simulations illustrate pulse splitting at a junction, and a controlled work–precision study compares members of the proposed family along the tested refinement paths.

The algorithm and exact-integration analysis extend to any fixed connected finite metric graph with standard continuity and Kirchhoff vertex conditions and uniform cable and gate laws (Remark 3). Section 2 presents the model, Section 3 defines the schemes and regularizations, Sections 4 and 5 establish stability and convergence, and Section 6 reports the numerical results. The appendix gives the smooth mask construction.

2. Mathematical Model

2.1. Domain and Geometry

We use the Y-shaped metric graph

$$\Gamma = A_0 \cup A_1 \cup A_2,$$

with one trunk and two daughter branches, shown in Figure 1. Its planar embedding in $\mathbb{R}^3$ is

$$\begin{aligned} A_0 &= \{(x, 0, 0) \ : \ -1 \le x \le 0\}, \\ A_1 &= \left\{ \left(\frac{s}{\sqrt{2}}, \frac{s}{\sqrt{2}}, 0 \right) \ : \ 0 \le s \le 1 \right\}, \\ A_2 &= \left\{ \left(\frac{s}{\sqrt{2}}, -\frac{s}{\sqrt{2}}, 0 \right) \ : \ 0 \le s \le 1 \right\}. \end{aligned}$$

Thus A_0 has arc-length coordinate $s = x \in [-1, 0]$, while A_1 and A_2 have arc-length coordinate $s \in [0, 1]$. The bifurcation node is located at $(0, 0, 0)$. The edgewise construction also applies to finite metric graphs with multiple junctions or cycles, using one shared degree of freedom at each vertex.

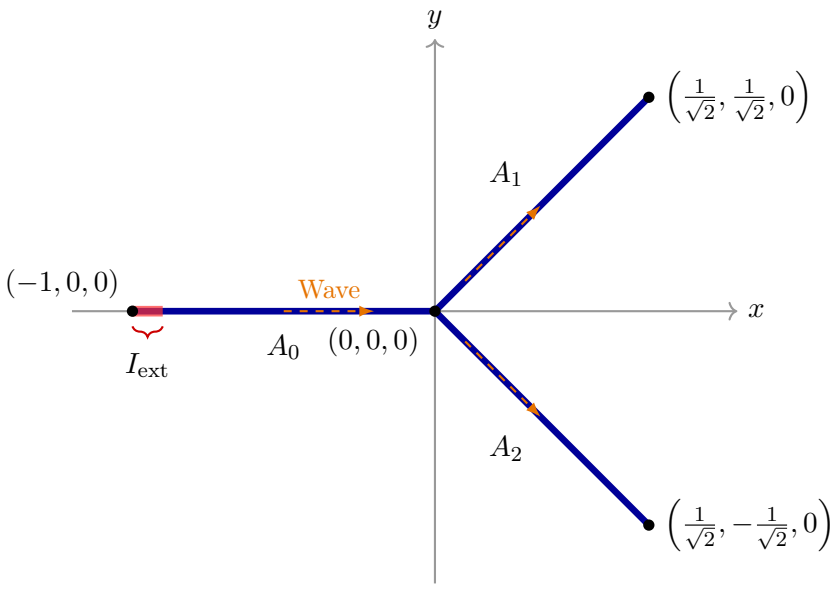


Figure 1: Planar representative of the branched axon graph $\Gamma = A_0 \cup A_1 \cup A_2$ embedded in $\mathbb{R}^3$. The external stimulus I_{ext} is applied near the left endpoint of the trunk.

2.2. Cable Equation and Hodgkin–Huxley Dynamics on the Graph

Let $V = (V_0, V_1, V_2)$ denote the transmembrane voltage, where each V_i is defined on the edge A_i. On each edge, the cable equation reads

$$C_m \frac{\partial V_i}{\partial t} = D \frac{\partial^2 V_i}{\partial s^2} - I_{\text{ion}}(V_i, m_i, h_i, n_i) + I_{\text{ext},i}(s,t), \qquad \text{on } A_i \times (0,T]. \tag{1}$$

where $D = a/(2R)$. Throughout the analysis we assume

$$C_m > 0, \qquad a > 0, \qquad R > 0, \qquad g_{\text{Na}} \geq 0, \qquad g_{\text{K}} \geq 0, \qquad g_{\text{L}} > 0,$$

so in particular $D > 0$. The ionic current is

$$I_{\text{ion}}(V, m, h, n) = g_{\text{Na}} m^3 h (V - E_{\text{Na}}) + g_{\text{K}} n^4 (V - E_{\text{K}}) + g_{\text{L}}(V - E_{\text{L}}). \tag{2}$$

Let $\mathcal{W} := \{m, h, n\}$. The gating variables $w \in \mathcal{W}$ satisfy

$$\frac{\partial w}{\partial t} = \alpha_w(V)(1 - w) - \beta_w(V) w = \alpha_w(V) - \lambda_w(V) w, \qquad \lambda_w(V) := \alpha_w(V) + \beta_w(V). \tag{3}$$

We use the classical Hodgkin–Huxley rate functions [1], in the shifted voltage convention where the resting potential is $V = 0$:

$$\alpha_m(V) = \frac{0.1(25 - V)}{\exp((25 - V)/10) - 1}, \qquad \beta_m(V) = 4 \exp(-V/18),$$

$$\alpha_h(V) = 0.07 \exp(-V/20), \qquad \beta_h(V) = \frac{1}{\exp((30 - V)/10) + 1},$$

$$\alpha_n(V) = \frac{0.01(10 - V)}{\exp((10 - V)/10) - 1}, \qquad \beta_n(V) = 0.125 \exp(-V/80).$$

Standard L'Hôpital regularizations are used at the removable singularities, so that $\alpha_m(25) = 1$ and $\alpha_n(10) = 0.1$. These rates are nonnegative and hence $\lambda_w(V) > 0$ for each gate $w \in \mathcal{W}$. Moreover, the interval $[0, 1]$ is positively invariant for each exact gate equation: at $w = 0$ the right-hand side is $\alpha_w(V) \geq 0$, while at $w = 1$ it is $-\beta_w(V) \leq 0$. At the outer endpoints we impose sealed-end conditions

$$\partial_s V_0(-1, t) = 0, \qquad \partial_s V_1(1, t) = 0, \qquad \partial_s V_2(1, t) = 0. \tag{4}$$

At the junction we impose continuity

$$V_0(0, t) = V_1(0, t) = V_2(0, t), \tag{5}$$

and Kirchhoff flux balance

$$D\,\partial_s V_0(0, t) = D\,\partial_s V_1(0, t) + D\,\partial_s V_2(0, t). \tag{6}$$

Here ∂_s denotes differentiation in the edge coordinates specified above; with this convention, (6) is the zero-net-flux condition at the junction.

2.3. Weak Formulation on the Metric Graph

Define

$$H^1(\Gamma) := \left\{ v = (v_0, v_1, v_2) \ : \ v_i \in H^1(A_i),\ v_0(0) = v_1(0) = v_2(0) \right\}.$$

For every integer $k \geq 1$, we use the graph Sobolev space

$$H^k(\Gamma) := \left\{ v \in H^1(\Gamma) \ : \ v_i \in H^k(A_i), \quad i = 0, 1, 2 \right\}, \qquad \|v\|^2_{H^k(\Gamma)} := \sum_{i=0}^{2} \|v_i\|^2_{H^k(A_i)}.$$

In particular, every element of $H^k(\Gamma)$ has a single common trace at the junction, while its higher derivatives are understood edgewise. The continuous problem is supplied with initial data

$$V(\cdot, 0) = V^0 \in H^1(\Gamma), \qquad w(\cdot, 0) = w^0 \in H^1(\Gamma), \quad w \in \mathcal{W}, \tag{7}$$

where $0 \leq w^0 \leq 1$ on Γ; compatibility required by the regularity assumptions in the convergence analysis is understood. For $u, v \in L^2(\Gamma)$, define

$$(u, v)_\Gamma := \sum_{i=0}^{2} \int_{A_i} u_i(s) v_i(s)\, ds, \qquad \|u\|^2_{L^2(\Gamma)} := (u, u)_\Gamma.$$

For brevity, we write $\|\cdot\| := \|\cdot\|_{L^2(\Gamma)}$ below.
The weak form of (1) is: find $V(t) \in H^1(\Gamma)$ such that

$$C_m(\partial_t V, v)_\Gamma + D(\partial_s V, \partial_s v)_\Gamma + (I_{\text{ion}}(V, m, h, n), v)_\Gamma = (I_{\text{ext}}, v)_\Gamma, \qquad \forall v \in H^1(\Gamma). \tag{8}$$

2.4. External Stimulus

For the analysis, I_{ext} is a prescribed source term satisfying

$$\sup_{0 \leq t \leq T} \|I_{\text{ext}}(\cdot, t)\| < \infty. \tag{9}$$

The compactly supported stimulus used in the computations is specified in Section 6.

3. Numerical Method

Throughout a computation, the independently selected degree $p \geq 1$ and BDF order $q \in \{1, \ldots, 5\}$ remain fixed, and the time step is constant. The finite element construction is edge-wise, with continuity imposed at graph vertices [28].

3.1. Finite Element Space and Nodal Norm

Let $\mathcal{T}_h$ be a partition of each edge into intervals, compatible at the graph vertices, and set $h := \max_{K \in \mathcal{T}_h} |K|$. Define

$$X_h^p := \{v_h \in H^1(\Gamma) : v_h|_K \in \mathbb{P}_p(K) \text{ for every } K \in \mathcal{T}_h\},$$

where $\mathbb{P}_p(K)$ is the space of polynomials of degree at most p. On the reference interval $[-1, 1]$, we use the $p+1$ Gauss–Lobatto nodes [37, Eq. (2.3.12)]

$$-1 = \xi_0 < \xi_1 < \cdots < \xi_p = 1.$$

For $p \geq 2$, the interior nodes are the zeros of L_p', where L_p is the Legendre polynomial normalized by $L_p(1) = 1$. The associated weights are

$$\omega_i^{(p)} = \frac{2}{p(p+1)L_p(\xi_i)^2} > 0, \qquad i = 0, \ldots, p.$$

Mapping these nodes affinely to each element gives the global nodal set $\mathcal{N}_h^p$. Nodes at a common element endpoint or graph vertex are identified. In particular, all incident branches share one junction degree of freedom.

Let $a_{K,i}$ denote the image of ξ_i on K. The assembled positive nodal weights are

$$M_a := \sum_{\substack{K \in \mathcal{T}_h,\ 0 \leq i \leq p \\ a_{K,i} = a}} \frac{|K|}{2} \omega_i^{(p)}, \qquad a \in \mathcal{N}_h^p.$$

Thus the junction weight includes contributions from every incident element. Since the Gauss–Lobatto rule is exact for constants, $\sum_{i=0}^p \omega_i^{(p)} = 2$ and $\sum_{a \in \mathcal{N}_h^p} M_a = |\Gamma|$. For real nodal vectors z and y, define

$$(z, y)_h := \sum_{a \in \mathcal{N}_h^p} M_a z(a) y(a), \tag{10}$$

and

$$\|z\|_h^2 := (z, z)_h. \tag{11}$$

We use the same notation for a nodal vector and its unique interpolant in X_h^p. This inner product will be used for the gate analysis; the voltage equation below uses the continuous graph inner product.

Lemma 1 (Equivalence of the nodal and continuous norms)**.** *For every integer $p \geq 1$,*

$$\|z_h\|^2 \leq \|z_h\|_h^2 \leq \left(2 + \frac{1}{p}\right) \|z_h\|^2, \qquad z_h \in X_h^p. \tag{12}$$

In particular, the equivalence constants can be chosen as 1 and 3, independently of h and p.

Proof. On the reference interval, write $\widehat{z} = \sum_{j=0}^{p} \widehat{z}_j L_j$. Legendre orthogonality and its Gauss–Lobatto discrete counterpart give [38]

$$\int_{-1}^{1} |\widehat{z}(\xi)|^2 \, d\xi = \sum_{j=0}^{p} \frac{2}{2j+1} |\widehat{z}_j|^2,$$
$$\sum_{i=0}^{p} \omega_i^{(p)} |\widehat{z}(\xi_i)|^2 = \sum_{j=0}^{p-1} \frac{2}{2j+1} |\widehat{z}_j|^2 + \frac{2}{p} |\widehat{z}_p|^2.$$

Only the highest-mode weight differs, by the factor $(2p+1)/p = 2+1/p \le 3$. Affine scaling to each element and assembly over the graph therefore prove (12). □

3.2. BDF Time Discretization

Let $\Delta t > 0$ be constant and $t_n = n\Delta t$. Following the BDF formulation in [35], define the newest-first BDF coefficients $a_j^{(q)}$ by

$$\sum_{j=0}^{q} a_j^{(q)} \zeta^j = \sum_{\ell=1}^{q} \frac{(1-\zeta)^\ell}{\ell}. \tag{13}$$

For a sequence $\{z^n\}$, the undivided BDF difference is

$$\delta_q z^{n+1} := \sum_{j=0}^{q} a_j^{(q)} z^{n+1-j}, \qquad n \ge q-1. \tag{14}$$

The approximation to the time derivative is $\delta_q z^{n+1}/\Delta t$. In particular, $a_0^{(q)} = \sum_{\ell=1}^{q} 1/\ell > 0$. The coefficients for the permitted orders are listed in Table 1.

Table 1: Newest-first coefficients of the undivided BDF difference.

q	$a_0^{(q)}$	$a_1^{(q)}$	$a_2^{(q)}$	$a_3^{(q)}$	$a_4^{(q)}$	$a_5^{(q)}$
1	1	-1	—	—	—	—
2	$\frac{3}{2}$	-2	$\frac{1}{2}$	—	—	—
3	$\frac{11}{6}$	-3	$\frac{3}{2}$	$-\frac{1}{3}$	—	—
4	$\frac{25}{12}$	-4	3	$-\frac{4}{3}$	$\frac{1}{4}$	—
5	$\frac{137}{60}$	-5	5	$-\frac{10}{3}$	$\frac{5}{4}$	$-\frac{1}{5}$

3.3. Gate Extrapolation and Mask

For each gate $w \in \mathcal{W}$, let $w_h^n \in X_h^p$ denote the numerical solution at t_n. For $n \ge q-1$, define the order-q extrapolation using the coefficients of the IMEX BDF polynomial in [35]:

$$w_*^{n+1} = \sum_{j=0}^{q-1} b_j^{(q)} w_h^{n-j}, \qquad b_j^{(q)} := (-1)^j \binom{q}{j+1}. \tag{15}$$

For example, $q = 3$ gives $w_*^{n+1} = 3w_h^n - 3w_h^{n-1} + w_h^{n-2}$. The extrapolated finite element gates need not remain in $[0, 1]$. To retain coercivity of the voltage equation, we use the fixed C^6 mask constructed in Appendix Appendix A. It satisfies

$$\phi(x) = x \quad (0 \le x \le 1), \qquad -\varepsilon \le \phi(x) \le 1 + \varepsilon, \qquad |\phi(x) - \phi(y)| \le L_\phi |x - y|, \qquad L_\phi = 1.$$

Choose $\varepsilon > 0$ so that

$$g_{\mathrm{Na}} \varepsilon (1 + \varepsilon)^3 \le \frac{g_{\mathrm{L}}}{2}.$$

These mask properties suffice for every permitted (q, p).

Write

$$\begin{aligned} &\boldsymbol{\mathcal{P}} = (m, h, n), \qquad \boldsymbol{\mathcal{P}}^n = (m(t_n), h(t_n), n(t_n)), \\ &\boldsymbol{\mathcal{P}}_h^n = (m_h^n, h_h^n, n_h^n), \qquad \boldsymbol{\mathcal{P}}_*^{n+1} = (m_*^{n+1}, h_*^{n+1}, n_*^{n+1}), \\ &\Phi(\mathbf{r}) := (\phi(r_m), \phi(r_h), \phi(r_n)), \qquad \mathbf{r} = (r_m, r_h, r_n). \end{aligned} \tag{16}$$

For any triple $\mathbf{r} = (r_m, r_h, r_n)$, define

$$\begin{aligned} G(\mathbf{r}) &= g_{\mathrm{Na}} r_m^3 r_h + g_{\mathrm{K}} r_n^4 + g_{\mathrm{L}}, \\ F(\mathbf{r}) &= g_{\mathrm{Na}} r_m^3 r_h E_{\mathrm{Na}} + g_{\mathrm{K}} r_n^4 E_{\mathrm{K}} + g_{\mathrm{L}} E_{\mathrm{L}}. \end{aligned} \tag{17}$$

These maps are evaluated pointwise on the masked extrapolated finite element fields $\Phi(\boldsymbol{\mathcal{P}}_*^{n+1})$.

3.4. Voltage Solve

For $n \ge q - 1$, given the voltage and gate histories at levels $n, n-1, \ldots, n-q+1$, define

$$\begin{aligned} G_h^{n+1} &:= G(\Phi(\boldsymbol{\mathcal{P}}_*^{n+1})), \\ F_h^{n+1} &:= F(\Phi(\boldsymbol{\mathcal{P}}_*^{n+1})), \\ f_h^{n+1} &:= F_h^{n+1} + I_{\mathrm{ext}}^{n+1}, \end{aligned} \tag{18}$$

where $I_{\mathrm{ext}}^{n+1} = I_{\mathrm{ext}}(\cdot, t_{n+1})$. Find $V_h^{n+1} \in X_h^p$ such that

$$\begin{aligned} \frac{C_m}{\Delta t} \big(\delta_q V_h^{n+1}, \chi_h\big)_\Gamma + D(\partial_s V_h^{n+1}, \partial_s \chi_h)_\Gamma + \big(G_h^{n+1} V_h^{n+1}, \chi_h\big)_\Gamma \\ = \big(f_h^{n+1}, \chi_h\big)_\Gamma, \qquad \chi_h \in X_h^p, \end{aligned} \tag{19}$$

All coefficients involving the gates are known from previous levels, so this equation is linear in V_h^{n+1}.

3.5. Voltage Cutoff for the Rate Functions

Let $M > 0$ be chosen larger than the L^∞-range of the exact voltage. Use a C^6 cutoff $\psi_M : \mathbb{R} \to \mathbb{R}$ satisfying

$$\psi_M(x) = x \quad \text{for } |x| \le M, \qquad |\psi_M(x)| \le M + 1, \qquad |\psi_M'(x)| \le 1.$$

The smooth saturation in Appendix Appendix A, applied symmetrically outside $[-M, M]$ with unit saturation width, provides such a cutoff. Using the removable-singularity regularizations of the rate functions, define

$$\widetilde{\alpha}_w(V) := \alpha_w(\psi_M(V)), \qquad \widetilde{\beta}_w(V) := \beta_w(\psi_M(V)), \qquad \widetilde{\lambda}_w(V) := \widetilde{\alpha}_w(V) + \widetilde{\beta}_w(V).$$

These functions are globally bounded and Lipschitz, and $\widetilde{\lambda}_w$ is strictly positive. If the extrapolated gate fields remain in $[0, 1]$ and the voltage arguments remain in $[-M, M]$, the mask and cutoff are inactive and the original Hodgkin–Huxley coefficients and rates are recovered.

3.6. Nodal Gate Update

After computing V_h^{n+1}, update each $w \in \mathcal{W}$ at every $a \in \mathcal{N}_h^p$ by

$$\frac{1}{\Delta t}\delta_q w_h^{n+1}(a) = \widetilde{\alpha}_w\big(V_h^{n+1}(a)\big) - \widetilde{\lambda}_w\big(V_h^{n+1}(a)\big) w_h^{n+1}(a), \qquad n \geq q-1. \tag{20}$$

Equivalently,

$$w_h^{n+1}(a) = \frac{-\sum_{j=1}^{q} a_j^{(q)} w_h^{n+1-j}(a) + \Delta t\, \widetilde{\alpha}_w\big(V_h^{n+1}(a)\big)}{a_0^{(q)} + \Delta t\, \widetilde{\lambda}_w\big(V_h^{n+1}(a)\big)}. \tag{21}$$

The denominator is positive for every $\Delta t > 0$. These nodal values, including the single value at the shared junction node, therefore determine a unique $w_h^{n+1} \in X_h^p$. Since V_h^{n+1} is already known, (21) is evaluated directly. Thus the complete timestep consists of one global linear voltage solve and independent scalar gate updates, with no nonlinear iteration.

Remark 1. *The gate update does not project the numerical values onto* $[0,1]$.

3.7. Initialization

Prescribe $V_h^j, w_h^j \in X_h^p$ for $j = 0, \ldots, q-1$ and $w \in \mathcal{W}$. The convergence analysis requires Assumption 2; the localized-stimulus experiment uses the exact resting histories (83).

4. Stability Analysis

We establish unconditional discrete-energy stability for the complete voltage–gate scheme, simultaneously for the BDF orders $q \in \{1, \ldots, 5\}$ and the conforming finite element degrees $p \geq 1$. Throughout this section the inner products in (19) are evaluated exactly. Neither estimate requires a numerical maximum principle or a restriction on the time step; dependence on the starting values is expressed through discrete history energies.

4.1. Basic Bounds Induced by the Mask

The pointwise mask bounds imply

$$\phi(m_*^{n+1})^3 \phi(h_*^{n+1}) \geq -\varepsilon(1+\varepsilon)^3.$$

And $\phi(n_*^{n+1})^4$ is nonnegative. Therefore the prescribed choice $g_{\mathrm{Na}}\varepsilon(1+\varepsilon)^3 \leq g_{\mathrm{L}}/2$ gives

$$\begin{aligned} 0 < g_0 := \frac{g_{\mathrm{L}}}{2} \leq G_h^{n+1} \leq g_1, \\ g_1 := g_{\mathrm{L}} + (g_{\mathrm{Na}} + g_{\mathrm{K}})(1+\varepsilon)^4, \qquad n \geq q-1. \end{aligned} \tag{22}$$

G_h^{n+1} is defined in (18). Similarly, from (17) and (18), we have

$$\left|F_h^{n+1}\right| \leq g_{\mathrm{Na}}(1+\varepsilon)^4 \left|E_{\mathrm{Na}}\right| + g_{\mathrm{K}}(1+\varepsilon)^4 \left|E_{\mathrm{K}}\right| + g_{\mathrm{L}} \left|E_{\mathrm{L}}\right| =: C_E.$$

Consequently,

$$\left\|F_h^{n+1}\right\| \leq |\Gamma|^{1/2} C_E. \tag{23}$$

Together with (9), this yields a constant C_F independent of h, Δt, q, p, and the numerical gate values, such that

$$\left\|f_h^{n+1}\right\| \leq C_F, \qquad n \geq q-1, \quad t_{n+1} \leq T. \tag{24}$$

In particular, these bounds require no control of the unmasked gate extrapolations.

4.2. BDF Multiplier Inequality

We use the admissible Nevanlinna–Odeh multipliers

$$(\eta_1, \eta_2, \eta_3, \eta_4, \eta_5) = (0, 0, 0.0836, 0.2879, 0.8160). \tag{25}$$

The multiplier construction follows from Dahlquist's G-stability theory and the Nevanlinna–Odeh criterion [39, 34, 35]. Precise values of the optimal multipliers are given in [36]. In particular, the optimal fourth-order value is approximately 0.287806557, so the exact decimal 0.2879 in (25) is chosen above that threshold.

Lemma 2 (BDF multiplier inequality)**.** *For each $q \in \{1, \dots, 5\}$, there is a symmetric positive-definite matrix $\mathbf{M}_q \in \mathbb{R}^{q\times q}$ such that every sequence in a real inner product space X satisfies*

$$\left(\delta_q z^{\,n+1}, z^{n+1} - \eta_q z^n\right)_X \geq \frac{1}{2}\left(\left\|\boldsymbol{\mathcal{Z}}_q^{n+1}\right\|_{\mathbf{M}_q,X}^2 - \left\|\boldsymbol{\mathcal{Z}}_q^{n}\right\|_{\mathbf{M}_q,X}^2\right), \qquad n \geq q-1, \tag{26}$$

where

$$\boldsymbol{\mathcal{Z}}_q^n := [z^n, z^{n-1}, \dots, z^{n-q+1}]^T, \qquad \left\|\boldsymbol{\mathcal{Z}}_q^n\right\|_{\mathbf{M}_q,X}^2 := \sum_{i,j=1}^{q} (M_q)_{ij} (z^{n+1-i}, z^{n+1-j})_X.$$

Moreover, there are constants $c_* > 0$ and $c^* < \infty$, independent of $q \in \{1, \dots, 5\}$ and of X, such that

$$c_* \sum_{j=0}^{q-1} \left\|z^{n-j}\right\|_X^2 \leq \left\|\boldsymbol{\mathcal{Z}}_q^n\right\|_{\mathbf{M}_q,X}^2 \leq c^* \sum_{j=0}^{q-1} \left\|z^{n-j}\right\|_X^2. \tag{27}$$

Proof. Apply the Nevanlinna–Odeh identity recalled by Akrivis [35, Lemma 1.1] and discard its nonnegative remainder. For the admissible multipliers in (25), let $\mathbf{G}_q$ denote the positive-definite matrix in that identity and let J_q reverse the history order. Taking $\mathbf{M}_q = 2J_q\mathbf{G}_qJ_q$ gives our newest-first convention and the factor $1/2$ in (26). The norm equivalence (27) follows from positive definiteness; its constants can be chosen independently of X and uniformly over the five orders $q = 1, \dots, 5$. □

We omit the subscript X when the underlying inner product is $(\cdot,\cdot)_\Gamma$.

4.3. Unconditional Voltage Stability

Theorem 1 (Masked-coefficient voltage stability)**.** *Fix $q \in \{1, \dots, 5\}$ and an integer $p \geq 1$. Let $\{V_h^n\} \subset X_h^p$ satisfy* (19)*, with starting values $V_h^0, \dots, V_h^{q-1} \in X_h^p$. Define*

$$\begin{aligned} \boldsymbol{\mathcal{V}}_{h,q}^n &:= [V_h^n, V_h^{n-1}, \dots, V_h^{n-q+1}]^T, \\ \mathcal{E}_V^n &:= \left\|\boldsymbol{\mathcal{V}}_{h,q}^n\right\|_{\mathbf{M}_q}^2 + \frac{D\eta_q^2\Delta t}{C_m}\left\|\partial_s V_h^n\right\|^2, \qquad n \geq q-1. \end{aligned} \tag{28}$$

There exists $C(T) > 0$, independent of h, Δt, p, and q, such that, for every $N \geq q-1$ with $t_N \leq T$,

$$\max_{q-1\leq m\leq N} \|V_h^m\|^2 \leq C(T)\left(\mathcal{E}_V^{q-1} + TC_F^2\right). \tag{29}$$

The same right-hand side, with a possibly enlarged $C(T)$, bounds $\max_{q-1\leq m\leq N} \mathcal{E}_V^m$.

Proof. For $n \geq q-1$, put

$$\chi_h^{n+1} := V_h^{n+1} - \eta_q V_h^n \in X_h^p.$$

Testing (19) with χ_h^{n+1} gives

$$\begin{aligned}\frac{C_m}{\Delta t}(\delta_q V_h^{n+1}, \chi_h^{n+1})_\Gamma + D(\partial_s V_h^{n+1}, \partial_s \chi_h^{n+1})_\Gamma \\ + (G_h^{n+1} V_h^{n+1}, \chi_h^{n+1})_\Gamma = (f_h^{n+1}, \chi_h^{n+1})_\Gamma.\end{aligned} \tag{30}$$

Step 1: the temporal and diffusion terms. Lemma 2 yields

$$(\delta_q V_h^{n+1}, \chi_h^{n+1})_\Gamma \geq \frac{1}{2}\left(\left\|\boldsymbol{\mathcal{V}}_{h,q}^{n+1}\right\|_{\mathbf{M}_q}^2 - \left\|\boldsymbol{\mathcal{V}}_{h,q}^{n}\right\|_{\mathbf{M}_q}^2\right). \tag{31}$$

The identity $2a(a-\eta_q b) = a^2 - \eta_q^2 b^2 + (a-\eta_q b)^2$ gives

$$D(\partial_s V_h^{n+1}, \partial_s \chi_h^{n+1})_\Gamma = \frac{D}{2}\left(\left\|\partial_s V_h^{n+1}\right\|^2 - \eta_q^2 \left\|\partial_s V_h^n\right\|^2 + \left\|\partial_s \chi_h^{n+1}\right\|^2\right). \tag{32}$$

Step 2: the reaction and source terms. The same identity, applied pointwise with the positive weight G_h^{n+1}, gives

$$\begin{aligned}(G_h^{n+1} V_h^{n+1}, \chi_h^{n+1})_\Gamma &= \frac{1}{2}\Big(\left\|\sqrt{G_h^{n+1}} V_h^{n+1}\right\|^2 - \eta_q^2 \left\|\sqrt{G_h^{n+1}} V_h^{n}\right\|^2 + \left\|\sqrt{G_h^{n+1}} \chi_h^{n+1}\right\|^2\Big) \\ &\geq \frac{g_0}{2}\left\|V_h^{n+1}\right\|^2 - \frac{\eta_q^2 g_1}{2}\left\|V_h^n\right\|^2 + \frac{g_0}{2}\left\|\chi_h^{n+1}\right\|^2.\end{aligned} \tag{33}$$

By Cauchy–Schwarz and Young's inequality,

$$(f_h^{n+1}, \chi_h^{n+1})_\Gamma \leq \frac{1}{2g_0}\left\|f_h^{n+1}\right\|^2 + \frac{g_0}{2}\left\|\chi_h^{n+1}\right\|^2. \tag{34}$$

The last term is absorbed by (33).

Step 3: the energy recurrence. Combining (30)–(34), multiplying by $2\Delta t/C_m$, and dropping the nonnegative term involving $\left\|\partial_s \chi_h^{n+1}\right\|^2$ yields

$$\begin{aligned}\left\|\boldsymbol{\mathcal{V}}_{h,q}^{n+1}\right\|_{\mathbf{M}_q}^2 - \left\|\boldsymbol{\mathcal{V}}_{h,q}^{n}\right\|_{\mathbf{M}_q}^2 + \frac{D\Delta t}{C_m}\left(\left\|\partial_s V_h^{n+1}\right\|^2 - \eta_q^2\left\|\partial_s V_h^n\right\|^2\right) + \frac{g_0 \Delta t}{C_m}\left\|V_h^{n+1}\right\|^2 \\ \leq \frac{\Delta t}{C_m g_0}\left\|f_h^{n+1}\right\|^2 + \frac{\eta_q^2 g_1 \Delta t}{C_m}\left\|V_h^n\right\|^2.\end{aligned} \tag{35}$$

By (27),

$$\left\|V_h^n\right\|^2 \leq c_*^{-1}\left\|\boldsymbol{\mathcal{V}}_{h,q}^n\right\|_{\mathbf{M}_q}^2 \leq c_*^{-1}\mathcal{E}_V^n.$$

The definition (28) therefore transforms (35) into

$$\begin{aligned}\mathcal{E}_V^{n+1} + \frac{D(1-\eta_q^2)\Delta t}{C_m}\left\|\partial_s V_h^{n+1}\right\|^2 + \frac{g_0 \Delta t}{C_m}\left\|V_h^{n+1}\right\|^2 \\ \leq (1 + C_v \Delta t)\mathcal{E}_V^n + \frac{\Delta t}{C_m g_0}\left\|f_h^{n+1}\right\|^2,\end{aligned} \tag{36}$$

where

$$C_v := \frac{\eta_q^2 g_1}{C_m c_*}.$$

For every order considered, $0 \le \eta_q < 1$, so both dissipation terms on the left are nonnegative. For $q = 1, 2$, $\eta_q = 0$ and $C_v = 0$; the history energy alone is sufficient, and the diffusion term remains nonnegative.

Step 4: iteration. Dropping the nonnegative dissipation terms in (36), iterating from $n = q-1$, and using $1 + C_v\Delta t \le \exp(C_v\Delta t)$ and (24) gives, for $q-1 \le m \le N$,

$$\begin{aligned}\mathcal{E}_V^m &\le \exp\big(C_v(t_m - t_{q-1})\big)\left(\mathcal{E}_V^{q-1} + \frac{1}{C_m g_0}\sum_{n=q-1}^{m-1}\Delta t\left\|f_h^{n+1}\right\|^2\right)\\ &\le \exp(C_v T)\left(\mathcal{E}_V^{q-1} + \frac{TC_F^2}{C_m g_0}\right). \end{aligned} \tag{37}$$

The empty sum for $m = q-1$ is zero. Since $c_*\left\|V_h^m\right\|^2 \le \mathcal{E}_V^m$, this proves (29). □

Remark 2. *The estimate is unconditional with respect to h and Δt. For $q = 3, 4, 5$, its starting energy contains $D\eta_q^2\Delta t\left\|\partial_s V_h^{q-1}\right\|^2/C_m$, so it is a discrete-energy estimate and does not assert a pure L^2-to-L^2 bound for arbitrary starting histories. For $q = 1, 2$ that additional term vanishes. Under the resting initialization* (83)*, all voltage starting values are zero and $\mathcal{E}_V^{q-1} = 0$; hence* (29) *reduces to* $\max_{q-1\le m\le N}\left\|V_h^m\right\|^2 \le C(T)TC_F^2$.

4.4. Unconditional Gate Stability

Theorem 2 (Gate stability)**.** *Fix $q \in \{1, \dots, 5\}$ and an integer $p \ge 1$. Let the voltage sequence be arbitrary, and let each gate sequence satisfy* (20) *with prescribed starting values $w_h^0, \dots, w_h^{q-1}$. For $n \ge q-1$, define*

$$\boldsymbol{\mathcal{W}}_{h,q}^n := [w_h^n, w_h^{n-1}, \dots, w_h^{n-q+1}]^T, \qquad \mathcal{E}_w^n := \left\|\boldsymbol{\mathcal{W}}_{h,q}^n\right\|_{\mathbf{M}_q,h}^2,$$

where the history norm is induced by $(\cdot,\cdot)_h$. There is a constant $C(T) > 0$, independent of h, Δt, p, and q, such that

$$\max_{q-1\le m\le N}\sum_{w\in\mathcal{W}}\mathcal{E}_w^m \le C(T)\left(\sum_{w\in\mathcal{W}}\mathcal{E}_w^{q-1} + T\right) \tag{38}$$

whenever $N \ge q-1$ and $t_N \le T$. Consequently, $\max_{q-1\le m\le N}\sum_{w\in\mathcal{W}}\left\|w_h^m\right\|_h^2$ *and the corresponding reconstructed $L^2(\Gamma)$ norms obey the same bound.*

Proof. The cutoff construction gives constant bounds, common to the three gates, such that

$$0 \le \widetilde{\alpha}_w(v) \le \alpha_1, \qquad 0 < \lambda_0 \le \widetilde{\lambda}_w(v) \le \lambda_1, \qquad v \in \mathbb{R}. \tag{39}$$

Fix $w \in \mathcal{W}$ and set

$$\zeta_h^{n+1} := w_h^{n+1} - \eta_q w_h^n, \quad \alpha_{w,h}^{n+1} := \widetilde{\alpha}_w(V_h^{n+1}), \quad \lambda_{w,h}^{n+1} := \widetilde{\lambda}_w(V_h^{n+1}).$$

Testing (20) with ζ_h^{n+1} in $(\cdot,\cdot)_h$, as Theorem 1, Lemma 2 and the weighted square identity give

$$\frac{1}{2\Delta t}\left(\mathcal{E}_w^{n+1}-\mathcal{E}_w^n\right)+\frac{\lambda_0}{2}\left\|w_h^{n+1}\right\|_h^2-\frac{\eta_q^2\lambda_1}{2}\left\|w_h^n\right\|_h^2 + \frac{\lambda_0}{2}\left\|\zeta_h^{n+1}\right\|_h^2 \le (\alpha_{w,h}^{n+1},\zeta_h^{n+1})_h.$$

Young's inequality and $\sum_{a\in\mathcal{N}_h^p} M_a = |\Gamma|$ yield

$$(\alpha_{w,h}^{n+1},\zeta_h^{n+1})_h \le \frac{|\Gamma|\alpha_1^2}{2\lambda_0}+\frac{\lambda_0}{2}\left\|\zeta_h^{n+1}\right\|_h^2.$$

After cancellation and use of (27),

$$\mathcal{E}_w^{n+1} \le (1+C_g\Delta t)\mathcal{E}_w^n+\frac{|\Gamma|\alpha_1^2}{\lambda_0}\Delta t, \qquad C_g := \frac{\eta_q^2\lambda_1}{c_*}. \tag{40}$$

Iterating (40) from $n = q-1$, using $1+C_g\Delta t \le \exp(C_g\Delta t)$, and summing over the three gates proves (38). The final assertions follow from (27) and Lemma 1. Uniformity over q and p follows from the fixed multiplier and nodal-norm constants. □

Combining Theorems 1 and 2 gives unconditional stability of the complete voltage–gate scheme:

$$\max_{q-1\le m\le N}\left(\mathcal{E}_V^m+\sum_{w\in\mathcal{W}}\mathcal{E}_w^m\right) \le C(T)\left(\mathcal{E}_V^{q-1}+\sum_{w\in\mathcal{W}}\mathcal{E}_w^{q-1}+T(1+C_F^2)\right).$$

5. Convergence Analysis

Fix a time order $q\in\{1,\dots,5\}$ and an integer polynomial degree $p\ge 1$. All time steps in this section are constant. We write I_h for the degree-p nodal interpolation operator into X_h^p and use the nodal norm (11) for gate errors, with the norm equivalence from Lemma 1.

For the exact gate $w\in\mathcal{W}$, define

$$w_I^n := I_h w(t_n), \qquad e_w^n := w_h^n - w_I^n. \tag{41}$$

Unless a different dependence is stated, constants below may depend on the graph, the physical parameters, the fixed regularizations, the regularity bounds, and p, q, but not on h or Δt.

Assumption 1 (Regularity and boundedness)**.** *The exact solution satisfies*

$$V, w \in W^{q+1,\infty}(0,T;H^{p+1}(\Gamma)), \qquad w\in\mathcal{W}.$$

The graph Sobolev norms are understood edgewise, with the common junction trace prescribed by $H^1(\Gamma)$. The exact initial gates satisfy $0\le w(\cdot,0)\le 1$. Moreover, there is a constant $M_0<M$ such that

$$|V(x,t)|\le M_0, \qquad (x,t)\in\Gamma\times[0,T].$$

The gate mask acts as the identity on the exact gates, and the voltage cutoff acts as the identity on the exact voltage. The regularized rates satisfy (39) and, for a constant L_M common to the three gates,

$$|\widetilde{\alpha}_w(u)-\widetilde{\alpha}_w(v)|+|\widetilde{\lambda}_w(u)-\widetilde{\lambda}_w(v)|\le L_M|u-v|, \qquad u,v\in\mathbb{R}. \tag{42}$$

5.1. Interpolation, extrapolation, and the Ritz projection

The standard degree-p interpolation estimate on each edge [40], summed over the graph, is

$$\|v - I_h v\| + h \|\partial_s (v - I_h v)\| \le C h^{p+1} \|v\|_{H^{p+1}(\Gamma)}. \tag{43}$$

The common junction trace ensures that $I_h v \in X_h^p$.

Lemma 3 (Extrapolation and coefficient consistency)**.** *Under Assumption 1, for each $w \in \mathcal{W}$ and $n \ge q-1$ with $t_{n+1} \le T$, the extrapolated gate satisfies*

$$\left\| w_*^{n+1} - w_I^{n+1} \right\|_h \le C \sum_{j=0}^{q-1} \left\| e_w^{n-j} \right\|_h + C\Delta t^q. \tag{44}$$

Here e_w^n is defined in (41)*. Let the exact coefficients at t_{n+1} be*

$$G^{n+1} := G(\boldsymbol{\mathcal{P}}^{n+1}), \qquad F^{n+1} := F(\boldsymbol{\mathcal{P}}^{n+1}). \tag{45}$$

With G_h^{n+1} and F_h^{n+1} defined in (18)*, we also have*

$$\left\| G_h^{n+1} - G^{n+1} \right\| + \left\| F_h^{n+1} - F^{n+1} \right\| \le C \sum_{w \in \mathcal{W}} \sum_{j=0}^{q-1} \left\| e_w^{n-j} \right\|_h + C(h^{p+1} + \Delta t^q). \tag{46}$$

Proof. Recall $b_j^{(q)} = (-1)^j \binom{q}{j+1}$ for $0 \le j \le q-1$. Polynomial extrapolation from the previous q time levels is exact for polynomials of degree at most $q-1$. Let $C^0(\Gamma)$ denote the space of continuous functions on Γ with the supremum norm. Taylor expansion at t_{n+1} gives, at every node $a \in \mathcal{N}_h^p$,

$$\left| \sum_{j=0}^{q-1} b_j^{(q)} w(a, t_{n-j}) - w(a, t_{n+1}) \right| \le C\Delta t^q \left\| \partial_t^q w \right\|_{L^\infty(t_{n+1-q}, t_{n+1}; C^0(\Gamma))}.$$

The right-hand side is bounded by $C\Delta t^q$ under Assumption 1, since the one-dimensional Sobolev embedding applies on every edge. The positive nodal weights sum to $|\Gamma|$, so the same estimate holds in $\|\cdot\|_h$. The identity

$$w_*^{n+1} - w_I^{n+1} = \sum_{j=0}^{q-1} b_j^{(q)} e_w^{n-j} + \sum_{j=0}^{q-1} b_j^{(q)} w_I^{n-j} - w_I^{n+1}$$

therefore proves (44).

Since $\phi(w(t_{n+1})) = w(t_{n+1})$, global Lipschitz continuity of ϕ, Lemma 1, and (43) give

$$\begin{aligned} \left\| \phi(w_*^{n+1}) - w(t_{n+1}) \right\| &\le L_\phi \left\| w_*^{n+1} - w(t_{n+1}) \right\| \\ &\le C \left\| w_*^{n+1} - w_I^{n+1} \right\|_h + C \left\| w_I^{n+1} - w(t_{n+1}) \right\| \\ &\le C \sum_{j=0}^{q-1} \left\| e_w^{n-j} \right\|_h + C(h^{p+1} + \Delta t^q). \end{aligned} \tag{47}$$

The masked gates lie in $[-\varepsilon, 1+\varepsilon]$. The function G and F are Lipschitz on this fixed compact box. Applying their pointwise Lipschitz bounds to (47) proves (46). This argument uses the exact gate bounds, not bounds on the polynomial interpolants between their nodes. □

Following the standard Ritz projection construction [40], define $R_h : H^1(\Gamma) \to X_h^p$ by

$$D(\partial_s(R_h v - v), \partial_s \chi_h)_\Gamma + (R_h v - v, \chi_h)_\Gamma = 0, \qquad \chi_h \in X_h^p. \tag{48}$$

The second term makes the bilinear form coercive.

Lemma 4 (Ritz projection on the metric graph)**.** *On a family of quasi-uniform edge meshes compatible at the junction,*

$$\|R_h v - v\| + h \|\partial_s(R_h v - v)\| \le C h^{p+1} \|v\|_{H^{p+1}(\Gamma)}, \qquad v \in H^{p+1}(\Gamma). \tag{49}$$

If $v \in W^{1,\infty}(0,T; H^{p+1}(\Gamma))$, then

$$\frac{1}{\Delta t} \left\|\delta_q (R_h v - v)^{n+1}\right\| \le C h^{p+1} \|\partial_t v\|_{L^\infty(t_{n+1-q}, t_{n+1}; H^{p+1}(\Gamma))}, \qquad n \ge q-1. \tag{50}$$

Proof. Let

$$a_\Gamma(u,v) := D(\partial_s u, \partial_s v)_\Gamma + (u,v)_\Gamma, \qquad \rho := R_h v - v.$$

Coercivity, Galerkin orthogonality, and (43) imply

$$\|\rho\|_{H^1(\Gamma)} \le C h^p \|v\|_{H^{p+1}(\Gamma)}.$$

For the corresponding metric-graph estimate with linear elements, see [28, Theorem 3.2]. For the L^2 estimate, let $\psi \in H^1(\Gamma)$ solve $a_\Gamma(\psi, \chi) = (\rho, \chi)_\Gamma$ for every $\chi \in H^1(\Gamma)$. Coercivity gives $\|\psi\|_{H^1(\Gamma)} \le C \|\rho\|$. On each edge, testing with compactly supported functions gives $-D\partial_s^2 \psi + \psi = \rho$ in the distributional sense. Hence $\partial_s^2 \psi \in L^2$ edgewise and

$$D \left\|\partial_s^2 \psi\right\| \le \|\psi\| + \|\rho\| \le C \|\rho\|, \qquad \|\psi\|_{H^2(\Gamma)} \le C \|\rho\|.$$

The weak formulation also enforces the natural sealed-end and Kirchhoff conditions. In particular, branching does not invalidate this edgewise regularity estimate. Since $p \ge 1$, interpolation of ψ gives $\|\psi - I_h \psi\|_{H^1(\Gamma)} \le C h \|\psi\|_{H^2(\Gamma)}$. Therefore

$$\|\rho\|^2 = a_\Gamma(\rho, \psi - I_h \psi) \le C h \|\rho\|_{H^1(\Gamma)} \|\psi\|_{H^2(\Gamma)} \le C h \|\rho\|_{H^1(\Gamma)} \|\rho\|.$$

This proves (49).

For the time-difference bound, R_h is linear and independent of time, so

$$\frac{1}{\Delta t} \delta_q (R_h v - v)^{n+1} = (R_h - I) \left(\frac{1}{\Delta t} \delta_q v^{n+1} \right).$$

The BDF coefficients satisfy $\sum_{j=0}^q a_j^{(q)} = 0$. Consequently,

$$\frac{1}{\Delta t} \delta_q v^{n+1} = \frac{1}{\Delta t} \sum_{j=1}^q a_j^{(q)} \big(v(t_{n+1-j}) - v(t_{n+1}) \big),$$

whose $H^{p+1}(\Gamma)$ norm is at most

$$\left(\sum_{j=1}^q j |a_j^{(q)}| \right) \|\partial_t v\|_{L^\infty(t_{n+1-q}, t_{n+1}; H^{p+1}(\Gamma))}.$$

Apply (49) to this quotient to obtain (50). □

Split the voltage error as

$$V_h^n - V(t_n) = \theta^n + \rho^n, \qquad \theta^n := V_h^n - R_h V(t_n), \qquad \rho^n := R_h V(t_n) - V(t_n). \tag{51}$$

The preceding lemma and Assumption 1 imply

$$\|\rho^n\| + h\,\|\partial_s \rho^n\| \le C h^{p+1}, \qquad \frac{1}{\Delta t}\left\|\delta_q \rho^{n+1}\right\| \le C h^{p+1}. \tag{52}$$

The first estimate holds at all time levels; the second holds whenever the full BDF history is available.

Assumption 2 (Starting errors)**.** *At the starting levels $j = 0, \dots, q-1$,*

$$\left\|V_h^j - V(t_j)\right\| + \sum_{w \in \mathcal{W}} \left\|w_h^j - I_h w(t_j)\right\|_h \le C_{\mathrm{init}}(h^{p+1} + \Delta t^q). \tag{53}$$

In addition,

$$\eta_q \Delta t^{1/2} \left\|\partial_s (V_h^{q-1} - R_h V(t_{q-1}))\right\| \le C_{\mathrm{init}}(h^{p+1} + \Delta t^q). \tag{54}$$

For $q = 1, 2$, the chosen multiplier has $\eta_q = 0$, so (54) imposes no additional condition. For the other orders it controls the gradient contribution to the initial discrete energy. It holds, for example, when $V_h^{q-1} = R_h V(t_{q-1})$.

In the one-step estimates and the main convergence theorem below, all integrals in the voltage equation are evaluated exactly.

5.2. One-step error estimates

Let $\mathbf{M}_q$ and η_q be the fixed multiplier matrix and parameter in Lemma 2. With θ^n and e_w^n defined in (51) and (41), respectively, define the newest-first history vectors for $n \ge q-1$ by

$$\mathbf{\Theta}^n = (\theta^n, \dots, \theta^{n-q+1})^T, \qquad \boldsymbol{E}_w^n = (e_w^n, \dots, e_w^{n-q+1})^T. \tag{55}$$

The history norms of the vectors in (55) are those of Lemma 2, using $(\cdot,\cdot)_\Gamma$ for $\mathbf{\Theta}^n$ and $(\cdot,\cdot)_h$ for $\boldsymbol{E}_w^n$. In both cases, (27) applies. Set

$$\begin{aligned} A_n &:= \|\mathbf{\Theta}^n\|_{\mathbf{M}_q}^2 + \frac{D\eta_q^2 \Delta t}{C_m}\,\|\partial_s \theta^n\|^2\,, \\ B_n &:= \sum_{w \in \mathcal{W}} \|\boldsymbol{E}_w^n\|_{\mathbf{M}_q,h}^2\,, \qquad r^2 := h^{2p+2} + \Delta t^{2q}. \end{aligned} \tag{56}$$

Lemma 5 (Voltage error estimate)**.** *Under Assumption 1, with A_n, B_n, and r defined in* (56)*,*

$$A_{n+1} \le (1 + C\Delta t) A_n + C\Delta t B_n + C\Delta t r^2, \qquad n \ge q-1, \tag{57}$$

for all steps with $t_{n+1} \le T$.

Proof. The BDF temporal residual of the exact voltage is

$$\tau_V^{n+1} := \frac{1}{\Delta t} \sum_{j=0}^{q} a_j^{(q)} V(t_{n+1-j}) - \partial_t V(t_{n+1}). \tag{58}$$

By the order-q consistency of the BDF formula, Taylor expansion with an integral remainder gives

$$\left\|\tau_V^{n+1}\right\| \le C\Delta t^q \left\|\partial_t^{q+1} V\right\|_{L^\infty(t_{n+1-q},t_{n+1};L^2(\Gamma))} \le C\Delta t^q. \tag{59}$$

Inserting (58) in the exact weak equation (8) and subtracting it from the exact-integral numerical equation (19) yields

$$\frac{C_m}{\Delta t}(\delta_q\theta^{n+1},\chi_h)_\Gamma + D(\partial_s\theta^{n+1},\partial_s\chi_h)_\Gamma + (G_h^{n+1}\theta^{n+1},\chi_h)_\Gamma = \mathcal{R}_V^{n+1}(\chi_h), \tag{60}$$

where G_h^{n+1} and F_h^{n+1} are defined in (17)-(18), G^{n+1} and F^{n+1} are defined in (45), and

$$\begin{aligned}\mathcal{R}_V^{n+1}(\chi_h) = & -\frac{C_m}{\Delta t}(\delta_q\rho^{n+1},\chi_h)_\Gamma - D(\partial_s\rho^{n+1},\partial_s\chi_h)_\Gamma - (G_h^{n+1}\rho^{n+1},\chi_h)_\Gamma \\ & - ((G_h^{n+1}-G^{n+1})V(t_{n+1}),\chi_h)_\Gamma + (F_h^{n+1}-F^{n+1},\chi_h)_\Gamma \\ & - C_m(\tau_V^{n+1},\chi_h)_\Gamma.\end{aligned}$$

Ritz orthogonality gives

$$D(\partial_s\rho^{n+1},\partial_s\chi_h)_\Gamma = -(\rho^{n+1},\chi_h)_\Gamma.$$

Using (52), (59), (46), the bounded exact voltage, and the uniform bound $G_h^{n+1} \le g_1$, we obtain

$$|\mathcal{R}_V^{n+1}(\chi_h)| \le C\left(h^{p+1} + \Delta t^q + \sum_{w\in\mathcal{W}}\sum_{j=0}^{q-1}\left\|e_w^{n-j}\right\|_h\right)\|\chi_h\|.$$

The history-norm equivalence in Lemma 2 and (56) therefore implies

$$|\mathcal{R}_V^{n+1}(\chi_h)| \le C(B_n + r^2)^{1/2}\|\chi_h\|. \tag{61}$$

By Young's inequality,

$$|\mathcal{R}_V^{n+1}(\chi_h)| \le \frac{g_0}{2}\|\chi_h\|^2 + C(B_n + r^2).$$

Test (60) with $\chi_h = \theta^{n+1} - \eta_q\theta^n$. The temporal, diffusion, and reaction estimates (31), (32), and (33) apply with V_h replaced by θ. After cancellation and multiplication by $2\Delta t/C_m$, the definitions in (56) give

$$\begin{aligned}A_{n+1} + \frac{D(1-\eta_q^2)\Delta t}{C_m}\left\|\partial_s\theta^{n+1}\right\|^2 + \frac{g_0\Delta t}{C_m}\left\|\theta^{n+1}\right\|^2 \\ \le A_n + \frac{\eta_q^2 g_1\Delta t}{C_m}\|\theta^n\|^2 + C\Delta t(B_n + r^2).\end{aligned}$$

Here a further nonnegative term containing $\|\partial_s\chi_h\|^2$ has been discarded. Since $\eta_q < 1$ and $\|\theta^n\|^2 \le CA_n$ by (27), dropping the remaining dissipation terms proves (57). □

Lemma 6 (Gate error estimate)**.** *Under Assumption 1, with A_n, B_n, and r defined in* (56)*,*

$$B_{n+1} \le (1 + C\Delta t)B_n + C\Delta t A_{n+1} + C\Delta t r^2, \qquad n \ge q-1, \tag{62}$$

for all steps with $t_{n+1} \le T$.

Proof. For each gate and each node $a \in \mathcal{N}_h^p$, define

$$\tau_w^{n+1}(a) := \frac{1}{\Delta t} \sum_{j=0}^{q} a_j^{(q)} w(a, t_{n+1-j}) - \partial_t w(a, t_{n+1}). \tag{63}$$

Taylor expansion and the BDF moment identities give

$$|\tau_w^{n+1}(a)| \le C\Delta t^q \left\| \partial_t^{q+1} w \right\|_{L^\infty(t_{n+1-q}, t_{n+1}; C^0(\Gamma))}.$$

Sobolev embedding, Assumption 1, and positivity of the nodal weights therefore yield

$$\left\| \tau_w^{n+1} \right\|_h \le C\Delta t^q. \tag{64}$$

Because the exact voltage is in the uncut range, the exact gate equation (3) uses the same regularized rates. We subtract nodal equation of (3) from the numerical gate equation (20) to obtain

$$\frac{1}{\Delta t} \delta_q e_w^{n+1} + \widetilde{\lambda}_w(V_h^{n+1}) e_w^{n+1} = R_w^{n+1}, \tag{65}$$

as an equality of nodal vectors, where e_w^{n+1} is defined in (41), and

$$\begin{aligned} R_w^{n+1} := {} & \widetilde{\alpha}_w(V_h^{n+1}) - \widetilde{\alpha}_w(I_h V(t_{n+1})) \\ & - \left[\widetilde{\lambda}_w(V_h^{n+1}) - \widetilde{\lambda}_w(I_h V(t_{n+1})) \right] w_I^{n+1} - \tau_w^{n+1}. \end{aligned} \tag{66}$$

All products and rate evaluations in this identity are nodal. Since $I_h V(t_{n+1})(a) = V(a, t_{n+1})$ and $0 \le w_I^{n+1}(a) \le 1$, (42) implies

$$\left\| R_w^{n+1} \right\|_h \le C \left\| V_h^{n+1} - I_h V(t_{n+1}) \right\|_h + C\Delta t^q. \tag{67}$$

Test (65) with $\zeta_w^{n+1} := e_w^{n+1} - \eta_q e_w^n$ in $(\cdot,\cdot)_h$, and set $\mathcal{E}_{e,w}^n := \|\boldsymbol{E}_w^n\|_{\mathbf{M}_q,h}^2$, where $\boldsymbol{E}_w^n$ is defined in (55). Lemma 2, the weighted square identity and bounds of $\widetilde{\lambda}_w$ in (39) give

$$\begin{aligned} \frac{1}{2\Delta t} \left(\mathcal{E}_{e,w}^{n+1} - \mathcal{E}_{e,w}^n \right) + \frac{\lambda_0}{2} \left\| e_w^{n+1} \right\|_h^2 - \frac{\eta_q^2 \lambda_1}{2} \left\| e_w^n \right\|_h^2 \\ + \frac{\lambda_0}{2} \left\| \zeta_w^{n+1} \right\|_h^2 \le (R_w^{n+1}, \zeta_w^{n+1})_h. \end{aligned}$$

By Young's inequality,

$$(R_w^{n+1}, \zeta_w^{n+1})_h \le \frac{1}{2\lambda_0} \left\| R_w^{n+1} \right\|_h^2 + \frac{\lambda_0}{2} \left\| \zeta_w^{n+1} \right\|_h^2.$$

After cancellation, multiplication by $2\Delta t$, and use of $c_* \left\| e_w^n \right\|_h^2 \le \mathcal{E}_{e,w}^n$ from (27) in Lemma 2, we obtain

$$\mathcal{E}_{e,w}^{n+1} \le (1 + C\Delta t) \mathcal{E}_{e,w}^n + C\Delta t \left\| R_w^{n+1} \right\|_h^2.$$

Summing over the gates and using (67) yields

$$B_{n+1} \le (1 + C\Delta t) B_n + C\Delta t \left\| V_h^{n+1} - I_h V(t_{n+1}) \right\|_h^2 + C\Delta t^{2q+1}. \tag{68}$$

To estimate the voltage term, write

$$V_h^{n+1} - I_h V(t_{n+1}) = \theta^{n+1} + R_h V(t_{n+1}) - I_h V(t_{n+1}).$$

Both terms on the right belong to X_h^p. Lemma 1, the Ritz estimate, and the interpolation estimate give

$$\|R_h V(t_{n+1}) - I_h V(t_{n+1})\|_h \le C \|R_h V(t_{n+1}) - I_h V(t_{n+1})\| \le C h^{p+1}.$$

Since $\left\|\theta^{n+1}\right\|^2 \le C A_{n+1}$, it follows that

$$\left\|V_h^{n+1} - I_h V(t_{n+1})\right\|_h^2 \le C A_{n+1} + C h^{2p+2}. \tag{69}$$

Substitution into (68) proves (62). □

5.3. Coupled convergence

Theorem 3 (Convergence under exact integration)**.** *Let $q \in \{1, \dots, 5\}$ and an integer $p \ge 1$ be fixed, and let the edge meshes be quasi-uniform and compatible at the junction. Assume that all integrals in* (19) *are evaluated exactly. Under Assumptions 1 and 2, there is a constant $C_{p,q}(T)$ independent of $h, \Delta t$ such that, for every constant step $\Delta t > 0$ and every integer $N \ge q - 1$ with $N\Delta t \le T$,*

$$\max_{0 \le n \le N} \left[\|V_h^n - V(t_n)\| + \sum_{w \in \mathcal{W}} \|w_h^n - I_h w(t_n)\|_h \right] \le C_{p,q}(T)(h^{p+1} + \Delta t^q). \tag{70}$$

The reconstructed gate functions also satisfy

$$\max_{0 \le n \le N} \sum_{w \in \mathcal{W}} \|w_h^n - w(t_n)\| \le C_{p,q}(T)(h^{p+1} + \Delta t^q). \tag{71}$$

No smallness condition on Δt and no relation between h and Δt are required.

Proof. If $N = q - 1$, (70) is the starting-error assumption. Otherwise, enlarge the constants in (57) and (62) to one common constant $C \ge 0$. With $x = C\Delta t$, the two estimates read

$$A_{n+1} \le (1 + x)A_n + xB_n + xr^2, \qquad B_{n+1} \le (1 + x)B_n + xA_{n+1} + xr^2.$$

The order in these recurrences matches the numerical algorithm: the voltage uses previous gates, and the new gates use the new voltage. Set $S_n := A_n + B_n$. Adding A_{n+1} to both sides of the second inequality, then bounding the right side A_{n+1} by the first inequality, gives

$$\begin{aligned} S_{n+1} &\le (1 + x)A_{n+1} + (1 + x)B_n + xr^2 \\ &\le (1 + x)^2 A_n + (1 + x)^2 B_n + (2x + x^2)r^2. \end{aligned}$$

Equivalently,

$$S_{n+1} + r^2 \le (1 + C\Delta t)^2 (S_n + r^2). \tag{72}$$

Iterating for $n = q - 1, \dots, m - 1$ and using $1 + C\Delta t \le e^{C\Delta t}$ yields

$$S_m + r^2 \le e^{2C(t_m - t_{q-1})}(S_{q-1} + r^2) \le e^{2CT}(S_{q-1} + r^2). \tag{73}$$

It remains to bound S_{q-1}. The voltage error split (51), the starting-error bound (53), and the Ritz estimate (52) give, at every starting level,

$$\|\theta^j\| \le \left\|V_h^j - V(t_j)\right\| + \|\rho^j\| \le C(h^{p+1} + \Delta t^q), \qquad 0 \le j \le q-1.$$

The upper history-norm bound (27) in Lemma 2, the starting gradient bound (54), and the definition of A_n in (56) therefore give

$$A_{q-1} \le C(h^{2p+2} + \Delta t^{2q}) = Cr^2.$$

Similarly, the nodal starting-error bound (53), the upper history-norm bound (27), and the definitions (41) and (56) give $B_{q-1} \le Cr^2$. Thus (73) proves

$$A_m + B_m \le C_{p,q}(T)r^2, \qquad q-1 \le m \le N.$$

The lower history-norm bound (27), the definitions (41) and (56) now imply

$$\|\theta^m\| + \sum_{w\in\mathcal{W}} \|e_w^m\|_h \le C_{p,q}(T)(h^{p+1} + \Delta t^q).$$

Adding the Ritz projection error bound (52) proves (70) at all computed levels, and (53) covers any earlier starting levels. Finally, for each gate,

$$\|w_h^n - w(t_n)\| \le C\,\|w_h^n - I_h w(t_n)\|_h + \|I_h w(t_n) - w(t_n)\|\,.$$

The nodal norm equivalence and (43) prove (71), including the case $N = q-1$. □

Remark 3 (Extension to finite metric graphs)**.** *The stability and convergence arguments also apply to a fixed connected metric graph with finitely many edges of positive length. Impose voltage continuity and Kirchhoff flux balance at each interior vertex, sealed-end conditions at degree-one vertices, and the same cable coefficients and gate kinetics on every edge. Use compatible quasi-uniform edge meshes and a single voltage and gate degree of freedom at each vertex. The assembled Gauss–Lobatto weights remain positive, the multiplier estimates are unchanged, and the dual Ritz problem has edge-wise H^2 regularity with the standard vertex conditions. Thus the proofs of Theorems 1, 2, and 3 carry over under the corresponding regularity and starting-value assumptions. Their constants may depend on the fixed graph, including its edge lengths and connectivity.*

5.4. Extension to numerical quadrature

Let the voltage equation now be evaluated by a numerical quadrature rule. At the computed solution, define its full exact-form defect by

$$\begin{aligned}\mathcal{Q}_h^{n+1}(\chi_h) := \frac{C_m}{\Delta t}(\delta_q V_h^{n+1}, \chi_h)_\Gamma + D(\partial_s V_h^{n+1}, \partial_s \chi_h)_\Gamma \\ + (G_h^{n+1} V_h^{n+1} - F_h^{n+1} - I_{\mathrm{ext}}^{n+1}, \chi_h)_\Gamma, \qquad \chi_h \in X_h^p,\end{aligned} \tag{74}$$

where every integral on the right is exact. Thus $\mathcal{Q}_h^{n+1}$ is the residual obtained when the quadrature-based solution is inserted into the exact-integral voltage equation.

Assumption 3 (Quadrature defect bound)**.** *There is a constant C_Q, independent of h and Δt, such that*

$$\mathfrak{q}_{n+1} := \sup_{\chi_h \in X_h^p,\ \chi_h \ne 0} \frac{|\mathcal{Q}_h^{n+1}(\chi_h)|}{\|\chi_h\|} \le C_Q h^{p+1}, \qquad n \ge q-1. \tag{75}$$

Corollary 1 (Convergence with numerical quadrature)**.** *Let the edge meshes be quasi-uniform and compatible at the junction, and use a constant time step as in Theorem 3. Under Assumptions 1, 2, and 3, the conclusions* (70)–(71) *hold for the quadrature-based scheme, with* $C_{p,q}(T)$ *also allowed to depend on* C_Q*.*

Proof. The voltage error equation (60) gains the additional term $\mathcal{Q}_h^{n+1}(\chi_h)$ on its right-hand side. Consequently,

$$|\mathcal{R}_V^{n+1}(\chi_h) + \mathcal{Q}_h^{n+1}(\chi_h)| \le \left[C(B_n + r^2)^{1/2} + \mathfrak{q}_{n+1}\right] \|\chi_h\| .$$

The same multiplier argument and Young's inequality therefore replace (57) by

$$A_{n+1} \le (1 + C\Delta t)A_n + C\Delta t B_n + C\Delta t \left(r^2 + \mathfrak{q}_{n+1}^2\right).$$

By (75), $\mathfrak{q}_{n+1}^2 \le C_Q^2 h^{2p+2} \le Cr^2$. Thus the original voltage recurrence is recovered with a modified constant. The gate recurrence is unchanged, so the proof of Theorem 3 applies verbatim. □

Remark 4. *Exact integration gives* $\mathcal{Q}_h^{n+1} = 0$*. For a numerical quadrature rule, verifying* (75) *requires a separate analysis.*

6. Numerical Experiments

Unless stated otherwise, the experiments use the Y-shaped graph in Figure 1, with three unit-length branches. Section 6.2 also considers a tree with two junctions and unequal edge lengths. All experiments use the physical parameters in Table 2. The computations use $p = 1, \dots, 4$. The gate mask and voltage cutoff use $\varepsilon = 10^{-3}$ and $M = 200$, respectively. The mask parameter satisfies $g_{\mathrm{Na}}\varepsilon(1+\varepsilon)^3 \le g_{\mathrm{L}}/2$. The forcing, starting histories, final times, and refinement choices are specified separately for each experiment below. At a grid time $t = t_n$, we write $V_h(t) = V_h^n$ and $w_h(t) = w_h^n$. All sampled times below lie on the corresponding time grids; the same convention applies to the numerical reference solutions on their own grids.

On each edge A_i of length L_i, let $\ell_i \in [0, L_i]$ denote arc length measured from the initial vertex of the oriented edge. For the unit-length Y-shaped graph, $L_i = 1$ and

$$\ell_0(s) = s + 1, \qquad \ell_1(s) = \ell_2(s) = s. \tag{76}$$

Table 2: Physical parameters shared by the numerical experiments.

Parameter	Value	Unit	Description
C_m	1.0	μF/cm^2	Specific membrane capacitance
a	0.1	μm	Axon radius
R	35.4	Ω cm	Intracellular resistivity
g_{Na}	120.0	mS/cm^2	Maximum Na^+ conductance
g_{K}	36.0	mS/cm^2	Maximum K^+ conductance
g_{L}	0.3	mS/cm^2	Maximum leak conductance
E_{Na}	115.0	mV	Na^+ reversal potential
E_{K}	-12.0	mV	K^+ reversal potential
E_{L}	10.5989...	mV	Leakage reversal potential

We choose $a = 0.1\,\mu$m, consistent in scale with thin parallel fibers [41]. The electrical parameters follow Hodgkin and Huxley [1], with E_{L} adjusted to enforce the exact resting equilibrium. Define the equilibrium gate functions by

$$w_\infty(V) := \frac{\alpha_w(V)}{\alpha_w(V) + \beta_w(V)}, \qquad w \in \mathcal{W}.$$

In all experiments, the leak reversal potential is chosen as

$$E_{\mathrm{L}}^* = \frac{g_{\mathrm{Na}} m_\infty(0)^3 h_\infty(0)(0 - E_{\mathrm{Na}}) + g_{\mathrm{K}} n_\infty(0)^4 (0 - E_{\mathrm{K}})}{g_{\mathrm{L}}} = 10.598920969\ldots \tag{77}$$

The computations use the double-precision value $E_{\mathrm{L}} = E_{\mathrm{L}}^*$; Table 2 displays only a rounded decimal. This choice makes $V = 0$ with $w = w_\infty(0)$ an exact equilibrium of the unforced model and supplies the resting histories in Section 6.3.

6.1. Hodgkin–Huxley Model with a Manufactured Solution

We first verify the numerical method using prescribed smooth voltage and gate fields, with source terms chosen so that these fields solve the coupled equations exactly. We consider the four pairs $(q, p) = (2, 1), (3, 2), (4, 3), (5, 4)$ in joint space–time refinement. Using (76), define

$$Q_i(s,t) := e^{-t} \cos\big(\kappa_i \ell_i(s)\big), \qquad \kappa_i := \frac{2\pi}{L_i}. \tag{78}$$

The manufactured solution is

$$\begin{aligned} V_i^{\mathrm{ex}}(s,t) &= 10 + 3Q_i(s,t), \\ m_i^{\mathrm{ex}}(s,t) &= 0.1 + 0.02Q_i(s,t), \\ h_i^{\mathrm{ex}}(s,t) &= 0.6 + 0.03Q_i(s,t), \\ n_i^{\mathrm{ex}}(s,t) &= 0.3 + 0.02Q_i(s,t). \end{aligned} \tag{79}$$

For $t \geq 0$, the gate mask and the voltage cutoff act as the identity on the exact fields. At every edge endpoint, $Q_i = e^{-t}$ and $\partial_s Q_i = 0$. Thus the fields have a common junction trace, and the voltage satisfies the sealed-end and Kirchhoff conditions. Figure 2 shows the same cosine voltage profile on each branch.

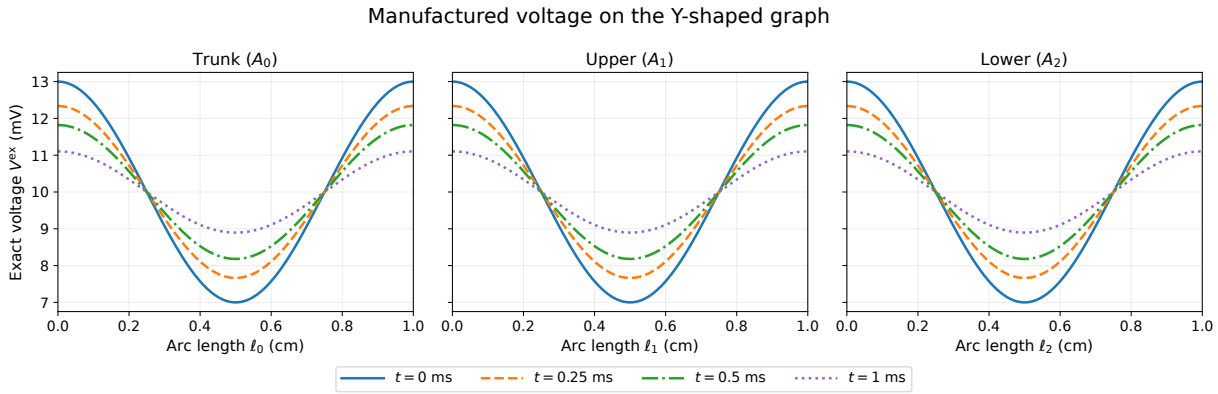


Figure 2: Exact manufactured voltage on the trunk and the two daughter branches.

For each order q, the starting levels use the manufactured histories:

$$V_h^j = R_h V^{\mathrm{ex}}(t_j), \qquad w_h^j = I_h w^{\mathrm{ex}}(t_j), \qquad j = 0, \dots, q-1, \quad w \in \mathcal{W},$$

with the Ritz projection defined in (48). Nonpolynomial integrals are evaluated using 24-point Gauss–Legendre quadrature on each element, and the continuous error norms use a 40-point rule.

Joint space–time refinement.. We set $T = 1$ and refine space and time together according to

$$h = \Delta t = \frac{1}{N_e}, \qquad N_e = 40, 80, 160, 320,$$

where N_e is the number of elements on each unit-length branch. Since $q = p + 1$, the bound $h^{p+1} + \Delta t^q$ predicts joint order q. For these tests, both the voltage and the reconstructed gates are compared directly with the analytic continuous fields:

$$\begin{aligned} E_V^{\mathrm{joint}} &:= \max_{t \in \mathcal{T}} \|V_h(t) - V^{\mathrm{ex}}(t)\|, \\ E_W^{\mathrm{joint}} &:= \max_{t \in \mathcal{T}} \max_{w \in \mathcal{W}} \|w_h(t) - w^{\mathrm{ex}}(t)\|, \\ \mathcal{T} &:= \{0, 0.05, \dots, 1\}. \end{aligned} \tag{80}$$

In particular, the joint gate error includes the continuous interpolation error. Rates are computed as $\log_2(E_{\mathrm{coarse}}/E_{\mathrm{fine}})$. Tables 3–5 report double-precision results with rates approaching the predicted orders. For BDF5–P_4, we use the `mpmath` library [42] with 50 decimal digits to prevent floating-point roundoff from obscuring fifth-order convergence, see Table 6.

Table 3: Joint refinement for BDF2–P_1 with $T = 1$ and $h = \Delta t$.

N_e	$h = \Delta t$	E_V^{joint}	Rate	E_W^{joint}	Rate
40	1/40	3.3888×10^{-3}	—	8.2715×10^{-5}	—
80	1/80	8.4534×10^{-4}	2.0032	2.0687×10^{-5}	1.9994
160	1/160	2.1122×10^{-4}	2.0008	5.1723×10^{-6}	1.9999
320	1/320	5.2798×10^{-5}	2.0002	1.2931×10^{-6}	2.0000

Table 4: Joint refinement for BDF3–P_2 with $T = 1$ and $h = \Delta t$.

N_e	$h = \Delta t$	E_V^{joint}	Rate	E_W^{joint}	Rate
40	1/40	8.4024×10^{-5}	—	8.1862×10^{-7}	—
80	1/80	1.1092×10^{-5}	2.9213	1.0235×10^{-7}	2.9996
160	1/160	1.4249×10^{-6}	2.9605	1.2795×10^{-8}	2.9999
320	1/320	1.8057×10^{-7}	2.9803	1.5994×10^{-9}	3.0000

6.2. Multiple Junctions and Unequal Edge Lengths

To check the assembly and spatial approximation beyond a single bifurcation, we use the five-edge tree in Figure 3. Its oriented edges are

$$A_0 = (v_0, v_1), \quad A_1 = (v_1, v_2), \quad A_2 = (v_1, v_3), \quad A_3 = (v_2, v_4), \quad A_4 = (v_2, v_5),$$

Table 5: Joint refinement for BDF4–P_3 with $T = 1$ and $h = \Delta t$.

N_e	$h = \Delta t$	E_V^{joint}	Rate	E_W^{joint}	Rate
40	1/40	1.7762×10^{-6}	—	1.4915×10^{-8}	—
80	1/80	1.2208×10^{-7}	3.8629	1.0371×10^{-9}	3.8462
160	1/160	8.0091×10^{-9}	3.9300	6.8419×10^{-11}	3.9220
320	1/320	5.1763×10^{-10}	3.9516	4.4372×10^{-12}	3.9467

Table 6: Joint refinement for BDF5–P_4 with $T = 1$ and $h = \Delta t$.

N_e	$h = \Delta t$	E_V^{joint}	Rate	E_W^{joint}	Rate
40	1/40	4.0277×10^{-8}	—	3.3569×10^{-10}	—
80	1/80	1.4300×10^{-9}	4.8158	1.2118×10^{-11}	4.7920
160	1/160	4.7645×10^{-11}	4.9076	4.0671×10^{-13}	4.8970
320	1/320	1.5375×10^{-12}	4.9537	1.3170×10^{-14}	4.9487

with metric lengths $(L_0, \ldots, L_4) = (1, 1.3, 1, \sqrt{2}, 1)$. Both v_1 and v_2 are junctions of degree three. A single global degree of freedom is shared by all incident edges at each junction; sealed conditions are imposed at the four leaves, and Kirchhoff balance is incorporated through the assembled weak formulation.

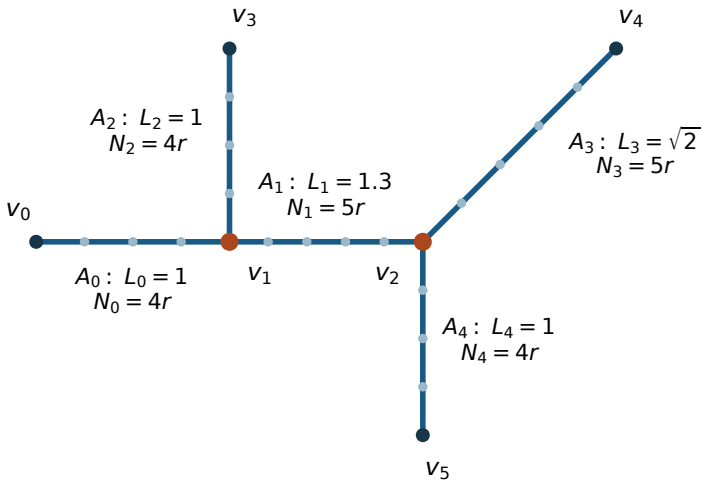


Figure 3: Tree with two junctions and unequal edge lengths. Orange vertices are junctions. Labels give the metric lengths and element counts; small light dots mark element boundaries for $r = 1$.

We reuse the manufactured fields in (78)-(79), with $\kappa_i = 2\pi/L_i$ on each edge. At every vertex these fields have the same trace and zero edgewise derivative, so they satisfy the vertex and leaf conditions. BDF5–P_4 is used with $T = 0.1$, fixed $\Delta t = 0.005$, and the manufactured starting histories specified in Section 6.1. The element counts are

$$(N_0, \ldots, N_4) = r(4, 5, 4, 5, 4), \qquad r = 1, 2, 4, 8,$$

so $h = \sqrt{2}/(5r)$ and the global number of degrees of freedom is $88r + 1$. Assembly uses 24-point Gauss–Legendre quadrature and continuous error evaluation uses a 40-point rule. At the sampled times $\mathcal{T}_g = \{0, 0.025, 0.05, 0.075, 0.1\}$, define

$$\begin{aligned} E_V &:= \max_{t\in\mathcal{T}_g} \|V_h(t) - V^{\mathrm{ex}}(t)\|\,, \\ E_W &:= \max_{t\in\mathcal{T}_g} \max_{w\in\mathcal{W}} \|w_h(t) - w^{\mathrm{ex}}(t)\|\,, \\ E_{V,1} &:= \max_{t\in\mathcal{T}_g} \|\partial_s(V_h(t) - V^{\mathrm{ex}}(t))\|\,. \end{aligned} \tag{81}$$

Here E_W includes gate interpolation error, and $E_{V,1}$ is the voltage H^1 seminorm error. Figure 4 shows L^2 rates approaching five for voltage and gates and an H^1 seminorm rate approaching four for voltage, as expected for P_4.

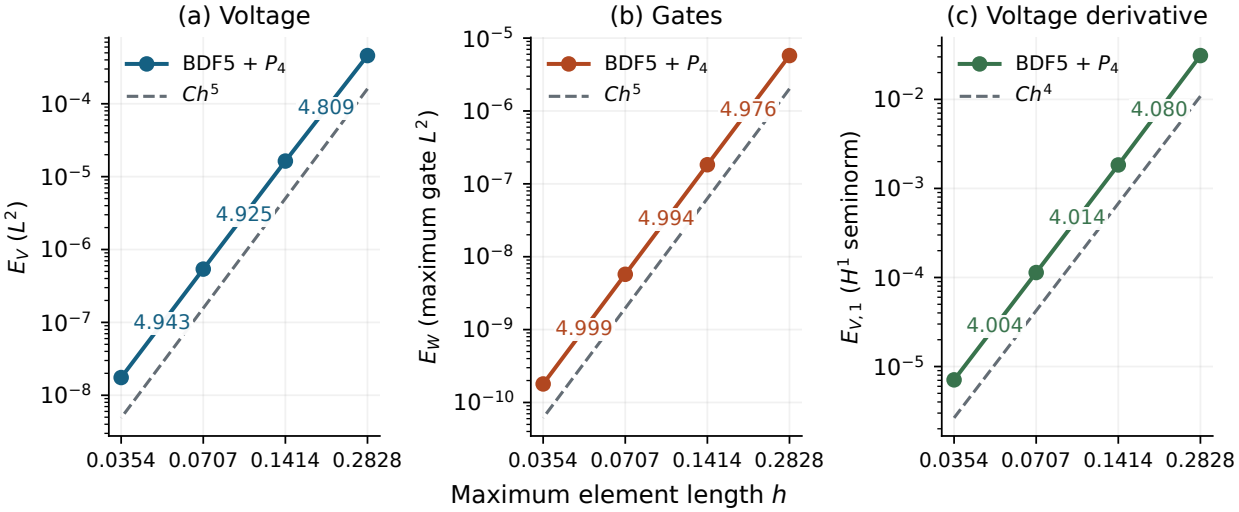


Figure 4: Spatial convergence on the two-junction tree with BDF5–P_4 and fixed $\Delta t = 0.005$: (a) voltage L^2 error, (b) maximum gate L^2 error, and (c) voltage derivative error (H^1 seminorm). Errors are maxima over $\mathcal{T}_g$. Dashed lines indicate fifth-order scaling in (a)–(b) and fourth-order scaling in (c); numbers give observed rates between successive meshes.

6.3. Hodgkin–Huxley Model with a Localized Stimulus

The external stimulus acts only on the trunk A_0, with amplitude $I_0 = 100.0$:

$$\begin{aligned} I_{\mathrm{ext},0}(s,t) &= I_0\,\phi_t(t)\phi_s(s), \qquad s \in [-1, 0], \\ I_{\mathrm{ext},1}(s,t) &= I_{\mathrm{ext},2}(s,t) = 0, \end{aligned}$$

$$\phi_t(t) = \begin{cases} e\exp\left(-\dfrac{0.01}{0.01 - (t-0.2)^2}\right), & t \in (0.1, 0.3), \\ 0, & \text{otherwise}, \end{cases}$$

$$\phi_s(s) = \begin{cases} e\exp\left(-\dfrac{0.01}{0.01 - (2s+1.9)^2}\right), & s \in (-1.0, -0.9), \\ 0, & \text{otherwise}. \end{cases} \tag{82}$$

This stimulus is smooth and compactly supported in time and space, so it satisfies (9).

We initialize at the resting equilibrium $V^0 = 0$, $w^0 = w_\infty(0)$. Since the stimulus starts at $t = 0.1\,\mathrm{ms}$ and all simulations satisfy $(q-1)\Delta t < 0.1\,\mathrm{ms}$, every starting level is an exact resting state:

$$V_h^j = 0, \qquad w_h^j = w_\infty(0), \qquad j = 0, \dots, q-1, \quad w \in \mathcal{W}. \tag{83}$$

These constants have zero starting and Ritz projection errors and satisfy Assumption 2. The corresponding resting gate values are

$$m_\infty(0) \approx 0.05293, \qquad h_\infty(0) \approx 0.59612, \qquad n_\infty(0) \approx 0.31768.$$

Voltage propagation.. We use a final time of $T = 76\,\mathrm{ms}$. Figure 5 shows the pulse traveling along the trunk and splitting symmetrically into the two daughter branches.

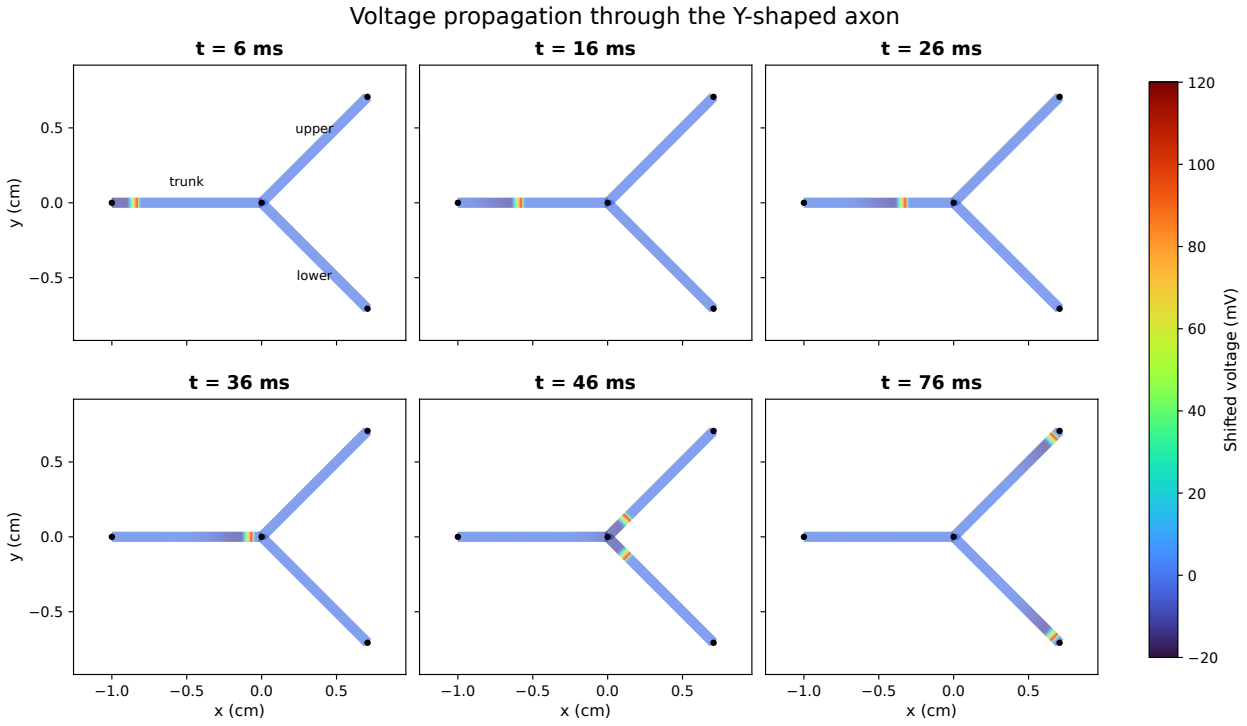


Figure 5: Voltage propagation under the localized stimulus, computed with BDF3–P_2, $h = 0.01\,\mathrm{cm}$, and $\Delta t = 0.02\,\mathrm{ms}$. The panels show $t = 6, 16, 26, 36, 46$, and $76\,\mathrm{ms}$, from left to right and top to bottom. The shared color scale gives the shifted voltage in mV.

Joint space–time convergence.. We also compare the physical Hodgkin–Huxley fields at $t_* = 46\,\mathrm{ms}$ under joint refinement, with $h = 1/N_e\,\mathrm{cm}$ and $\Delta t = 1/N_e\,\mathrm{ms}$. All comparisons use the same BDF5–P_4 numerical reference with 1280 elements per branch and $\Delta t_{\mathrm{ref}} = 1/1280\,\mathrm{ms}$. Assembly uses 24-point Gauss–Legendre quadrature. Define the continuous graph errors

$$\begin{aligned} E_V^{46} &:= \|V_h(t_*) - V_{\mathrm{ref}}(t_*)\|\,, \\ E_W^{46} &:= \max_{w \in \mathcal{W}} \|w_h(t_*) - w_{\mathrm{ref}}(t_*)\|\,, \\ E_{V,H^1}^{46} &:= \left[(E_V^{46})^2 + \left\|\partial_s\big(V_h(t_*) - V_{\mathrm{ref}}(t_*)\big)\right\|^2\right]^{1/2}. \end{aligned} \tag{84}$$

Table 7 reports these errors and the rates $\log_2(E_{N_e}/E_{2N_e})$. The finest BDF2–P_1 L^2 rates are close to two, while the BDF4–P_3 and BDF5–P_4 rates approach four and five. The BDF3–P_2 L^2 slopes remain above three on the available grids. The finest voltage H^1 rates for P_2, P_3, and P_4 are close to two, three, and four, respectively; the P_1 rate has not yet settled to one. These are comparisons with a finite numerical reference; above-nominal slopes do not establish a higher formal order. Figure 6 illustrates the approach of the BDF2–P_1 pulse profiles to the reference under refinement.

Table 7: Joint refinement for the physical Hodgkin–Huxley problem at $t = 46\,\mathrm{ms}$.

N_e	E_V^{46}	Rate	E_W^{46}	Rate	E_{V,H^1}^{46}	Rate
			BDF2–P_1			
80	3.9162×10^{1}	—	4.4531×10^{-1}	—	1.7855×10^{3}	—
160	2.8259×10^{1}	0.4707	2.9816×10^{-1}	0.5787	1.8869×10^{3}	−0.0797
320	9.3142×10^{0}	1.6012	9.3383×10^{-2}	1.6749	1.1639×10^{3}	0.6971
640	2.3901×10^{0}	1.9623	2.3940×10^{-2}	1.9637	3.4881×10^{2}	1.7384
1280	5.9685×10^{-1}	2.0017	5.9793×10^{-3}	2.0014	9.5080×10^{1}	1.8752
			BDF3–P_2			
80	2.1457×10^{1}	—	2.2041×10^{-1}	—	1.7926×10^{3}	—
160	2.3469×10^{0}	3.1926	2.3495×10^{-2}	3.2298	3.4564×10^{2}	2.3747
320	1.6159×10^{-1}	3.8603	1.6167×10^{-3}	3.8612	3.0136×10^{1}	3.5197
640	1.0437×10^{-2}	3.9526	1.0447×10^{-4}	3.9519	5.0169×10^{0}	2.5866
1280	6.7427×10^{-4}	3.9522	6.7640×10^{-6}	3.9490	1.2063×10^{0}	2.0562
2560	4.5786×10^{-5}	3.8803	4.6276×10^{-7}	3.8695	3.0174×10^{-1}	1.9993
			BDF4–P_3			
80	4.5711×10^{0}	—	4.5705×10^{-2}	—	6.4449×10^{2}	—
160	1.2888×10^{-1}	5.1484	1.2884×10^{-3}	5.1487	2.1321×10^{1}	4.9178
320	2.4644×10^{-3}	5.7087	2.4902×10^{-5}	5.6932	2.0798×10^{0}	3.3578
640	6.2218×10^{-5}	5.3077	6.5536×10^{-7}	5.2478	2.8153×10^{-1}	2.8851
1280	3.0331×10^{-6}	4.3585	3.3021×10^{-8}	4.3108	3.5440×10^{-2}	2.9898
			BDF5–P_4			
80	7.1823×10^{-1}	—	7.1537×10^{-3}	—	1.0780×10^{2}	—
160	6.7239×10^{-3}	6.7390	6.7835×10^{-5}	6.7205	4.0406×10^{0}	4.7376
320	7.6875×10^{-5}	6.4507	7.8463×10^{-7}	6.4339	2.9298×10^{-1}	3.7857
640	2.0972×10^{-6}	5.1960	2.3119×10^{-8}	5.0849	1.6429×10^{-2}	4.1565

6.4. Runtime Benchmark

We compare computational work at a prescribed solution error, following the accuracy–cost perspective used in electrophysiological time-integration studies [20, 21]. Here the comparison is between members of the same BDF–FEM family under common implementation and reference-solution settings. At $T = 46\,\mathrm{ms}$, we compare accuracy and computational cost for the same problem as in Section 6.3, using the same physical parameters and resting histories. The pairs BDF2–P_1, BDF3–P_2, and BDF4–P_3 use $N_e = 80, 160, 320, 640, 1280$ elements per branch; BDF5–P_4 uses

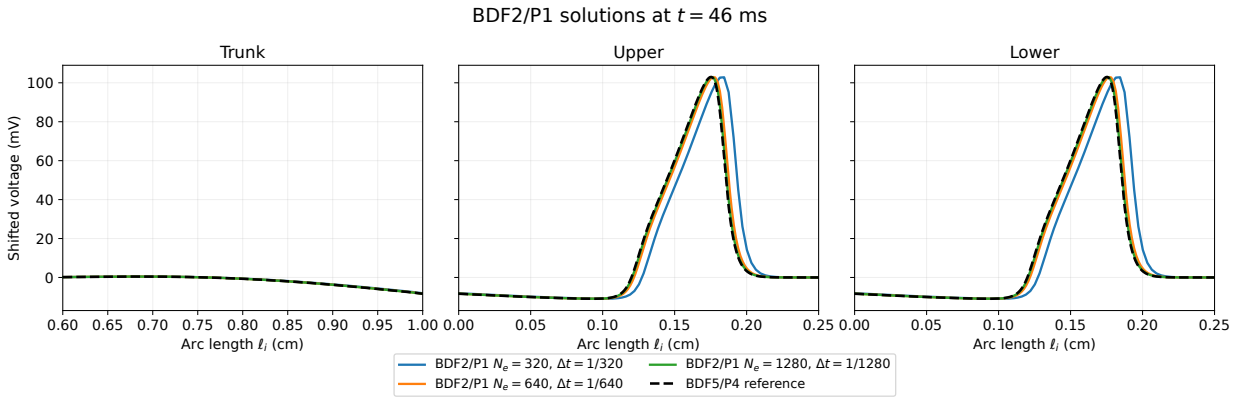


Figure 6: Zoomed BDF2–P_1 voltage profiles at $t = 46\,\text{ms}$ for $N_e = 320, 640, 1280$, with $h = 1/N_e\,\text{cm}$ and $\Delta t = 1/N_e\,\text{ms}$. The dashed curve is the common BDF5–P_4 reference used in Table 7.

$N_e = 80, 160, 320, 640$. In every case, $h = 1/N_e\,\text{cm}$ and $\Delta t = 1/N_e\,\text{ms}$. All cases use an optimized evaluation of the gate mask ϕ from Appendix Appendix A. If all extrapolated gate values at the assembly quadrature points lie in $[0, 1]$, we return a copy of these values directly, since $\phi(x) = x$ there. This avoids evaluating the polynomial and hyperbolic-tangent saturation branches. If any value lies outside $[0, 1]$, the full mask is evaluated on the entire array. This shortcut preserves the mask values and the numerical scheme. Assembly uses 24-point Gauss–Legendre quadrature. The errors in (84) are evaluated against the common BDF5–P_4 reference with $N_e = 1280$ and $\Delta t_{\text{ref}} = 1/1280\,\text{ms}$, using 8-point quadrature on the union of the candidate and reference element boundaries.

Each of the 19 cases is computed three times in fresh serial processes, with an identical untimed warm-up and randomized case order within each repetition block. All 57 runs use one numerical-library thread on an Intel Core i5-12400, with Python 3.13.13, NumPy 2.4.4, and SciPy 1.17.1. Elapsed wall-clock time includes mesh construction, solver setup, starting histories, and time integration. Imports, warm-up, reference loading, file output, and error evaluation are excluded. Figure 7 plots the arithmetic mean of the three runtimes against the final-time errors; horizontal bars show the minimum and maximum runtimes.

For voltage $L^2(\Gamma)$ error $E_V^{46} \le 10^{-3}$, the fastest tested cases are listed in Table 8. BDF2–P_1 does not reach this target on the tested grids. BDF5–P_4 has the lowest mean runtime at this target among the tested joint refinement paths. All errors are relative to the same finite numerical reference.

Table 8: Fastest tested cases with $E_V^{46} \le 10^{-3}$.

Scheme	N_e	E_V^{46}	Mean elapsed time (s)
BDF3–P_2	1280	6.7427×10^{-4}	546.7
BDF4–P_3	640	6.2218×10^{-5}	170.6
BDF5–P_4	320	7.6875×10^{-5}	48.6

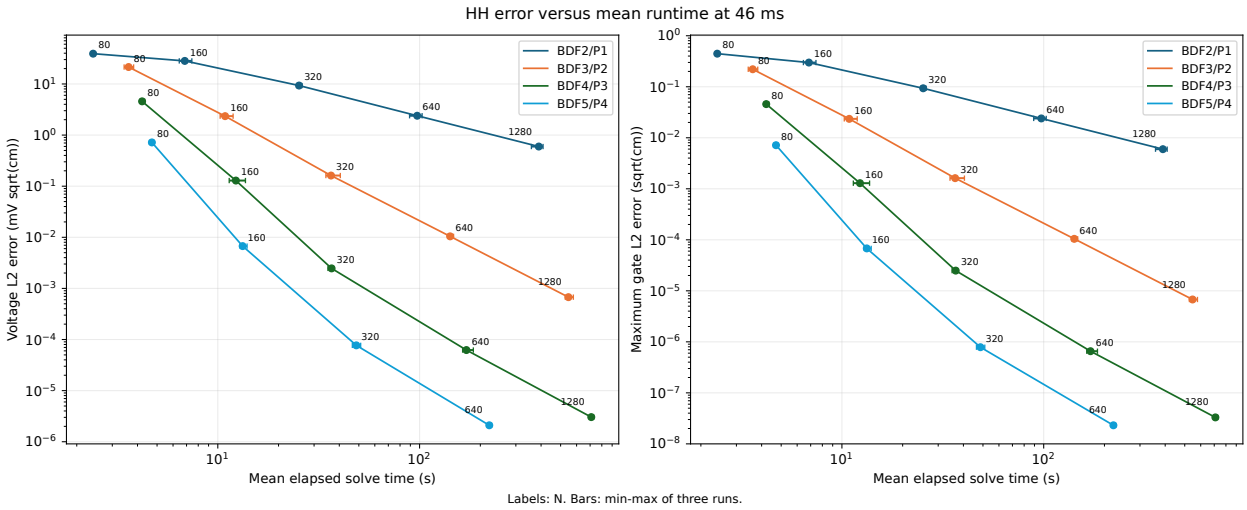


Figure 7: Final-time $L^2(\Gamma)$ errors versus mean elapsed runtime at $T = 46\,\mathrm{ms}$: voltage (left) and maximum gate error (right). Point labels give N_e; horizontal bars show the range of three runs.

7. Conclusions

We have developed and analyzed a linearly implicit IMEX BDF–FEM family for the Hodgkin–Huxley cable system on a branched metric graph, using constant time steps, orders $q = 1, \ldots, 5$, and any fixed spatial degree $p \geq 1$. Smooth masking of the extrapolated gates preserves voltage-equation coercivity without projecting the numerical gates, and a voltage cutoff gives bounded Lipschitz gate rates. Each timestep requires one linear voltage solve followed by independent nodal gate updates. Under exact spatial integration, the complete regularized scheme satisfies unconditional bounds in discrete history energies. Under regularity and starting-history assumptions, the voltage and reconstructed gates converge in $L^2(\Gamma)$ at the optimal rate $O(h^{p+1} + \Delta t^q)$, without a timestep restriction or an h–Δt coupling.

Manufactured solutions recover the predicted orders through BDF5–P_4 on the Y-shaped graph, and an additional two-junction test verifies assembly and spatial accuracy with unequal edge lengths. Physical simulations demonstrate pulse propagation and splitting on the Y-shaped axon. In the controlled work–precision comparison within the proposed family, higher-order variants attain the tested accuracy target at lower runtime, with BDF5–P_4 fastest among the tested cases. These comparisons use a common finite numerical reference and the tested joint-refinement paths.

For numerical quadrature, the convergence rate remains conditional on the stated bound on the full exact-form residual. The Gauss–Legendre implementation exhibits the expected rates, while a rule-specific proof of this defect bound for the masked coefficients and stimulus remains open. Such an analysis should identify practical quadrature orders that preserve the theoretical accuracy uniformly under refinement.

Under the stated assumptions, the exact-integration analysis extends to any fixed connected finite metric graph with standard continuity and Kirchhoff conditions, sealed outer endpoints, and uniform cable and gate laws. The experiments use uniform coefficients and smooth data; physical pulse propagation is tested on a symmetric Y-shaped graph, while the two-junction manufactured fields have zero edgewise derivatives at every vertex. Testing pulses through successive junctions and extending the method to heterogeneous networks are natural next steps. Spatial and temporal

adaptivity could improve efficiency when regularity varies, but variable time stepping requires further stability and convergence analysis. Solver and preconditioning studies should also address scalability on larger neuronal graphs.

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Appendix A. The $\mathcal{C}^{2N}$ Continuous Masking Function

For an integer $N \geq 1$, define the truncated Maclaurin series of $\operatorname{arctanh}(u)$ by

$$P_N(u) = \sum_{k=1}^{N} \frac{u^{2k-1}}{2k-1}. \tag{A.1}$$

For $\varepsilon > 0$, set

$$\phi(x) = \begin{cases} \varepsilon \tanh\left(P_N\left(\dfrac{x}{\varepsilon}\right)\right), & \text{if } x < 0, \\ x, & \text{if } 0 \leq x \leq 1, \\ 1 + \varepsilon \tanh\left(P_N\left(\dfrac{x-1}{\varepsilon}\right)\right), & \text{if } x > 1. \end{cases} \tag{A.2}$$

This function equals the identity on $[0, 1]$ and takes values in $[-\varepsilon, 1 + \varepsilon]$.

Appendix A.1. Smoothness

Since $P_N(u)$ agrees with $\operatorname{arctanh}(u)$ through degree $2N - 1$, we have

$$\tanh(P_N(u)) = u + \mathcal{O}(u^{2N+1}) \qquad \text{as } u \to 0.$$

Thus the Taylor series around each patching point matches the identity function through order $2N$, and the piecewise function is $\mathcal{C}^{2N}$.

Appendix A.2. Global Lipschitz Constant

On either outer branch, the derivative has the form

$$\phi'(x) = \frac{P_N'(u)}{\cosh^2(P_N(u))}, \qquad P_N'(u) = \sum_{k=0}^{N-1} u^{2k} > 0,$$

where $u = x/\varepsilon$ or $u = (x-1)/\varepsilon$. Since $P_N(u) - \operatorname{arctanh}(u) = \mathcal{O}(u^{2N+1})$ and $(\cosh^2)'(z) = \sinh(2z) = \mathcal{O}(z)$ near zero,

$$\cosh^2(P_N(u)) = \frac{1}{1-u^2} + \mathcal{O}(u^{2N+2}) \qquad \text{as } u \to 0.$$

The entire function $\cosh^2(P_N(u))$ has a globally convergent even Taylor series with nonnegative coefficients: P_N has nonnegative odd coefficients, and $\cosh^2 z$ has nonnegative even coefficients. The local expansion identifies the coefficients through degree $2N - 2$ as one. Therefore, for every real u,

$$\cosh^2(P_N(u)) = \sum_{k=0}^{N-1} u^{2k} + \sum_{k=N}^{\infty} c_k u^{2k} \geq P_N'(u), \qquad c_k \geq 0.$$

Thus $0 < \phi'(x) \leq 1$ globally. Since $\phi'(x) = 1$ on $(0, 1)$, the optimal Lipschitz constant is $L_\phi = 1$.

Appendix A.3. The C^6 Mask Used in the Computations

The choice $N = 3$ gives

$$P_3(u) = u + \frac{u^3}{3} + \frac{u^5}{5}. \tag{A.3}$$

The preceding arguments establish $\phi \in \mathcal{C}^6(\mathbb{R})$ and $L_\phi = 1$. More precisely, as $x \to 0^-$,

$$\varepsilon \tanh\left(P_3\left(\frac{x}{\varepsilon}\right)\right) = x - \frac{x^7}{7\varepsilon^6} + \mathcal{O}\left(\frac{x^9}{\varepsilon^8}\right).$$

The nonzero seventh-order term prevents $\mathcal{C}^7$ matching with the identity at $x = 0$; the same conclusion holds at $x = 1$ by translation. Figure A.8 shows this mask for the value of ε used in the experiments.

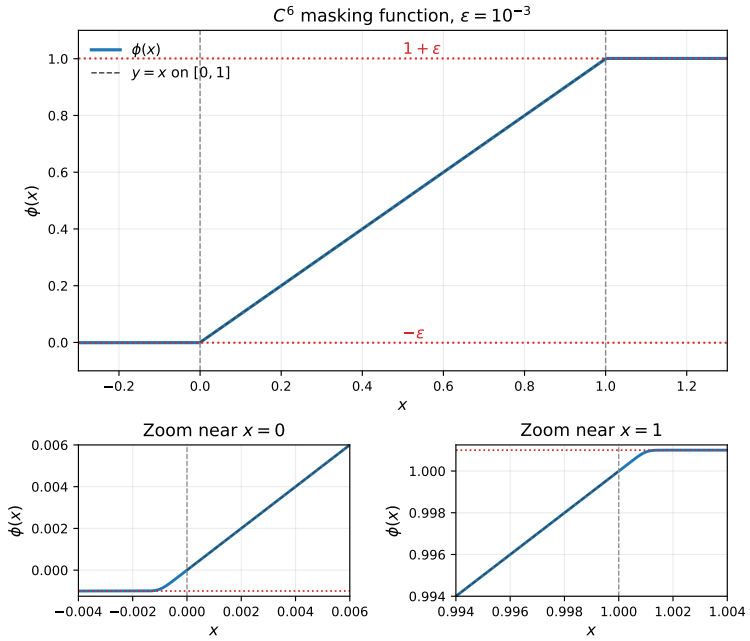


Figure A.8: The $\mathcal{C}^6$ masking function with $N = 3$ and $\varepsilon = 10^{-3}$, including details near the patching points.